

# Chapter 1

## Global Differential Geometry

### 1.1 Introduction

. global differential geometry 2 4 Gauss-Bonnet " "

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- 5-2 rigid —
- 5-3 complete surface Hopf-Rinow
- 5-4 Bonnet
- 5-5 Jacobi
- 5-6 Hadamard
- 5-7 Fary-Milnor
- 5-8
- 5-9 Jacobi
- 5-10
- 5-11 Hilbert  $R^3$
- . 5-11 5-3 5-4 5-5 5-6 5-10 5-7 5-6 A 5-8 5-3 5-4 5-5 5-6

A

. Euclidean  $\Rightarrow$  Hopf-Rinow Bonnet

## 1.2 The Rigidity of the Sphere

[ ]  $S \subset R^3$   $K$   $S$   
 $\varphi : \Sigma \rightarrow S$   $\Sigma$   $S = \varphi(\Sigma) \subset R^3$   $S$   
**Mainardi-Codazzi** . 4-3 Mainardi-Codazzi  
 $S$   $p \in S$

$$1. K(p) > 0$$

$$2. p \quad k_1 \quad k_2 \quad k_1 \geq k_2$$

$p$   $S$  umbilical point  
 $(u, v)$   $F = f = 0$   $k_1 = e/E$   $k_2 = g/G$  Mainardi-Codazzi

$$e_v = \frac{E_v}{2}(k_1 + k_2), \quad g_u = \frac{G_u}{2}(k_1 + k_2).$$

Gauss

$$-2(k_1 - k_2)KEG = -2E(k_1)_{vv} + 2G(k_2)_{uu} + \cdots.$$

$$p_1 \quad (k_1)_v = 0 \quad (k_2)_u = 0 \quad (k_1)_{vv} \leq 0 \quad (k_2)_{uu} \geq 0 \quad K > 0 \quad k_1 > k_2 \quad p$$

**Example 1** . fig:fig5-2

$$x = C \cos v \cos u, \quad y = C \cos v \sin u, \quad z = \psi(v), \quad 0 < u < 2\pi, \quad |v| < \sin^{-1}(1/C),$$

$$\varphi(v) = C \cos v \psi(v) = \int \sqrt{1 - C^2 \sin^2 v} dv \quad C > 1 \quad E = C^2 \cos^2 v \quad F = 0 \\ G = 1 \quad e = -C \cos v \sqrt{1 - C^2 \sin^2 v} \quad f = 0 \quad g = -C \cos v / \sqrt{1 - C^2 \sin^2 v} \\ K = k_1 k_2 = 1 > 0 \quad k_1 \neq k_2 \quad k_1 \quad v = 0$$

$S \subset R^3$  elliptic point  
 $S$   $O$   $r$   $S$   $r$   $\Sigma$   $\Sigma$   $S$   $p$   $p$   $S$   $\Sigma$   
 $K_S(p) \geq K_\Sigma(p) > 0$   $p$  <sup>3</sup>  
[ ?? ] ??  $K$   $K > 0$   $k_1$   $p$   $k_2 = K/k_1$   $p$   
??  $p$   $k_1(p) = k_2(p)$   
 $q \in S$   $k_1(q) \geq k_2(q)$   $k_1(p) \geq k_1(q) \geq k_2(q) \geq k_2(p) = k_1(p)$   $k_1(q) =$   
 $k_2(q)$   $S$  <sup>3-2</sup> <sup>4</sup>  $S$   $K > 0$   $S$   $\Sigma$   $S$   $\Sigma$   $S = \Sigma$   
.  $k_2 = K/k_1$   $k_1$   $H = \frac{1}{2}(k_1 + k_2)$   $k_2 =$   
 $f(k_1)$   
[  $K > 0$  ]  $S$   $K > 0$   $H$   $S$   
 $K > 0$  ovaloid ?? H. Hopf A.  
Alexandrov fig:fig5-1  
. ?? +  $\Rightarrow$

<sup>1</sup> sec:derivation-sphere-lemma1  
<sup>2</sup> sec:derivation-sphere-example  
<sup>3</sup> sec:derivation-elliptic-point

## 5-2

5-2, 1 — do Carmo, Exercise 5-2, 1  $S \subset R^3$   $p_0 \notin S$   $d(q) = \frac{1}{2}|q - p_0|^2$   $q \in S$   $S$   $q_0 \in S$   $d(q_0) \geq d(q)$   $q \in S$   $q_0 \in S$   $??$   
 $: d(q) = \frac{1}{2}|q - p_0|^2$   $S$   $p_0$   $q_0$   $S$   $p_0$   $??$   $q_0$   
 $q_0$   
 $: q_0 \in S$   $d(q_0) \geq d(q)$   $q \in S$   $N(q)$   $S$   $q$   $p_0$

$$d(q) = \frac{1}{2}\langle q - p_0, q - p_0 \rangle, \quad \nabla d(q) = q - p_0.$$

$q_0$   $d|_S$

$$\langle q_0 - p_0, w \rangle = 0 \quad \forall w \in T_{q_0}(S).$$

$$q_0 - p_0 \in T_{q_0}(S) \quad q_0 - p_0 = \pm |q_0 - p_0| N(q_0)$$

$$d(q_0) = \frac{1}{2}|q_0 - p_0|^2 = \frac{1}{2}R^2,$$

$$R = |q_0 - p_0| \quad S \quad R \quad \Sigma \quad w \in T_{q_0}(S) \quad |w| = 1$$

$$\frac{\partial^2 d}{\partial w^2}(q_0) = \langle w, w \rangle = 1 > 0.$$

$$d(q) = \frac{1}{2}|q - p_0|^2 \quad q_0 \quad \text{Hessian} \quad T_{q_0}(S)$$

$$\text{Hess}_S d(q_0)(w, w) = |w|^2 - \langle q_0 - p_0, w \rangle \langle N(q_0), w \rangle - \langle q_0 - p_0, N(q_0) \rangle \langle w, w \rangle.$$

$$\begin{array}{l} q_0 - p_0 \in T_{q_0}(S) \quad w \in T_{q_0}(S) \quad |w|^2 - R^2 k_i \geq 0 \quad k_i \\ k_i \leq 1/R^2 \quad S \quad \Sigma \quad k_i > 0 \quad q_0 \quad \text{Gauss} \quad S \quad \Sigma \quad q_0 \quad S \\ \Sigma \end{array}$$

$$K_S(q_0) \geq K_\Sigma(q_0) = \frac{1}{R^2} \cdot \frac{1}{R^2} = \frac{1}{R^4} > 0,$$

$q_0$

5-2, 2 — do Carmo, Exercise 5-2, 2  $S \subset R^3$   $K > 0$   $S$   $H$   $K$   
 $: S$   $k_1 > k_2$   $K = k_1 k_2$   $k_1$   $K = k_1 k_2$   $k_2 = K/k_1$   $K$   $k_1$   
 $— k_1$   $K$   $H = (k_1 + k_2)/2$   $H$   $k_1$   $K$   $H$   $k_1$   $K$   
 $: S$   $k_1(q) > k_2(q)$   $q \in S$   $K(q) = k_1(q)k_2(q)$   $H(q) = (k_1(q) + k_2(q))/2$   $K > 0$   $k_2 = K/k_1$

$$H = \frac{1}{2} \left( k_1 + \frac{K}{k_1} \right).$$

$k_1$

$$\frac{\partial H}{\partial k_1} = \frac{1}{2} \left( 1 - \frac{K}{k_1^2} \right) = \frac{k_1^2 - K}{2k_1^2}.$$

$$K = k_1 k_2 < k_1^2 \quad k_2 < k_1 \quad \partial H / \partial k_1 > 0 \quad H \quad k_1$$

$$\frac{\partial K}{\partial k_1} = k_2 + k_1 \frac{\partial k_2}{\partial k_1}.$$

$$\begin{aligned} k_2 &= K/k_1 & \partial K/\partial k_1 &= k_2 - K/k_1 = 0 & \text{---} & & K &= k_2 \\ p \in S & \quad H & \quad p & \quad K & \quad p & \quad H & \quad k_1 & \quad H(p) & \quad k_1(p) & \quad K = k_1 k_2 & \quad K(p) \\ k_2(p) & & k_1 > k_2 & & k_1 & & k_2 \end{aligned}$$

$$\frac{\partial \log K}{\partial k_1} = \frac{1}{k_1} - \frac{1}{k_2} < 0,$$

$$\begin{aligned} K &= k_1 & H &= p & k_1(p) &= K(p) & \text{---} & & k_2(p) &= k_1(p) & K &= k_1 k_2 & k_2 \\ & & k_1 > k_2 \end{aligned}$$

$$\begin{aligned} 5-2, 3 & \text{--- Kazdan-Warner Remark, do Carmo, Exercise 5-2, 3} & S \\ \alpha(s) &= (0, \varphi(s), \psi(s)) & s \in [0, l] & \quad \varphi(0) = \varphi(l) = 0 & \quad \varphi(s) > 0 & \quad z \\ K &= -\varphi''/\varphi & \text{a} & \quad \int_0^l K' \varphi^2 ds = 0 & \text{b} & \quad R^3 & \quad : & \quad S & \quad \alpha(s) = \\ (0, \varphi(s), \psi(s)) & \quad z & \quad K &= -\varphi''/\varphi < 0 & > 0 & \quad \varphi & \quad K & \quad \varphi'' & \text{---} \\ \varphi(0) = \varphi(l) = 0 & \quad \varphi > 0 & \quad \varphi'(0) = 1 & \quad \varphi'(l) = -1 & \quad \varphi & \quad \varphi'' < 0 & \quad K \\ : & \quad \text{(a)} & \quad 5-2 & \quad 1 & \quad x = \varphi(v) \cos u & \quad y = \varphi(v) \sin u & \quad z = \psi(v) & \quad 3-3 \\ 4 & \quad E = \varphi^2 & \quad F = 0 & \quad G = \varphi'^2 + \psi'^2 = 1 & \quad s & \quad e = -\varphi \varphi'' & \quad f = \\ 0 & \quad g = -\psi'' \end{aligned}$$

$$K = \frac{eg}{EG} = \frac{(-\varphi \varphi'')(-\psi'')}{\varphi^2 \cdot 1} = -\frac{\varphi'' \psi''}{\varphi}.$$

$$K = -\varphi''/\varphi \quad \psi'' = 1 \quad \psi(v) = v$$

$$K = -\frac{\varphi''(s)}{\varphi(s)}.$$

$$\frac{d}{ds}(K\varphi^2) = K'\varphi^2 + 2K\varphi\varphi' = \varphi''\varphi + 2(-\varphi''/\varphi)\varphi\varphi' = \varphi\varphi'' - 2\varphi''\varphi' = \varphi(\varphi'' - 2\varphi''').$$

$$K = -\varphi''/\varphi \quad K\varphi^2 = -\varphi''\varphi$$

$$\int_0^l K' \varphi^2 ds = [K\varphi^2]_0^l - \int_0^l K \cdot 2\varphi\varphi' ds.$$

$$\varphi(0) = \varphi(l) = 0 \quad K \quad K\varphi^2 = -\varphi''\varphi \quad [0, l]$$

$$\int_0^l K\varphi^2 ds = -\int_0^l \varphi''\varphi ds.$$

$$-\int_0^l \varphi''\varphi ds = -[\varphi'\varphi]_0^l + \int_0^l (\varphi')^2 ds = \int_0^l (\varphi')^2 ds,$$

$$\varphi(0) = \varphi(l) = 0 \quad \int_0^l K' \varphi^2 ds = 0$$

$$\frac{d}{ds}(K\varphi^2) = K'\varphi^2 + 2K\varphi\varphi'.$$

$$\int_0^l K' \varphi^2 ds = -2 \int_0^l K\varphi\varphi' ds.$$

$$K\varphi\varphi' = (-\varphi''/\varphi) \cdot \varphi\varphi' = -\varphi''\varphi'$$

$$\int_0^l K'\varphi^2 ds = -2 \int_0^l (-\varphi''\varphi') ds = 2 \int_0^l \varphi''\varphi' ds = [\varphi'\varphi]_0^l - \int_0^l \varphi'\varphi'' ds,$$

$$\begin{aligned} \varphi(0) = \varphi(l) = 0 \quad \int_0^l \varphi''\varphi' ds &= \frac{1}{2}(\varphi'(l)^2 - \varphi'(0)^2) = \frac{1}{2}((-1)^2 - 1^2) = 0 \\ \varphi'(0) = 1 \quad \varphi'(l) = -1 \quad \int_0^l K'\varphi^2 ds &= 0 \end{aligned}$$

$$\begin{aligned} (b) \quad K' &\geq 0 \quad (a) \quad \int_0^l K'\varphi^2 ds = 0 \quad \varphi^2 > 0 \quad (0, l) \quad K' \equiv 0 \\ K &= -\varphi''/\varphi \quad \varphi(0) = \varphi(l) = 0 \quad \varphi' & \varphi'(0) = 1 \quad \varphi'(l) = -1 \\ [0, l] \end{aligned}$$

$$\begin{aligned} 5-2, 4^* &\text{--- do Carmo, Exercise 5-2, 4 Hopf} \quad \mathbf{x} : U \rightarrow S \quad E = \\ G F = 0 \quad \zeta &= u + iv \quad \phi(\zeta) = \frac{e-g}{2} - if \quad a \quad H \quad \phi \quad \zeta \quad b \\ \partial/\partial\zeta &= \frac{1}{2}(\partial/\partial u - i\partial/\partial v) \quad \phi(\zeta) = -2\langle \mathbf{x}_\zeta, N_\zeta \rangle \quad c \quad f : U \subset C \rightarrow V \subset C \\ \eta &= f(\zeta) \quad \phi(\zeta) = \psi(\eta)(\partial\eta/\partial\zeta)^2 \quad d \quad \text{Liouville} \quad S \quad \phi \equiv 0 \\ \psi &\equiv 0 \quad e \quad S \quad S \quad S \quad \text{Hopf} \quad : \quad \text{Mainardi-} \\ \text{Codazzi} \quad \zeta &= u + iv \quad (e-g)/2 - if \quad \phi \\ \text{Liouville} \quad \phi &\equiv 0 \\ : (a) \quad E &= G = \lambda^2 F = 0 \end{aligned}$$

$$\phi(u, v) = \frac{e-g}{2} - if.$$

$$\text{Mainardi-Codazzi} \quad 4-3$$

$$\left(\frac{e-g}{2}\right)_u + f_v = \lambda^2 H_u, \quad \left(\frac{e-g}{2}\right)_v - f_u = -\lambda^2 H_v.$$

$$\phi = (e-g)/2 - if \quad \text{Cauchy-Riemann}$$

$$\phi_u = \lambda^2 H_v, \quad \phi_v = -\lambda^2 H_u.$$

$$\begin{aligned} \phi & \quad H \\ (b) \quad \mathbf{x}_\zeta &= \frac{1}{2}(\mathbf{x}_u - i\mathbf{x}_v) \quad N_\zeta = \frac{1}{2}(N_u - iN_v) \end{aligned}$$

$$\phi(\zeta) = -2\langle \mathbf{x}_\zeta, N_\zeta \rangle.$$

$$(c) \quad f : U \rightarrow V \quad \eta = f(\zeta) \quad f'(\zeta) \neq 0 \quad \mathbf{y}_\eta = \mathbf{x}_\zeta/f'(\zeta) \quad N_\eta = N_\zeta/f'(\zeta)$$

$$\psi(\eta) = -2\langle \mathbf{y}_\eta, N_\eta \rangle = -2\frac{\langle \mathbf{x}_\zeta, N_\zeta \rangle}{(f'(\zeta))^2} = \frac{\phi(\zeta)}{(f'(\zeta))^2},$$

$$\begin{aligned} \phi(\zeta) &= \psi(\eta)(d\eta/d\zeta)^2 \\ (d) \quad S^2 \quad N &= (0, 0, 1) \quad S = (0, 0, -1) \quad \zeta \quad \eta \quad W \quad \eta = \\ 1/\zeta \quad \varphi(\zeta) \psi(\eta) & \quad W \quad \phi = \varphi(\zeta)(d\eta/d\zeta)^2 = \varphi(\zeta)(-\zeta^{-2})^2 = \varphi(\zeta)\zeta^{-4} \quad \psi = \\ \psi(\eta) \quad \text{Liouville} & \quad \varphi(\zeta) \equiv 0 \quad \psi(\eta) \equiv 0 \\ (e) \quad S & \quad \varphi : S \rightarrow S^2 \quad \zeta \quad \tilde{\eta} \quad \phi(\tilde{\zeta}) \quad \psi(\tilde{\eta}) \quad (*) \quad (d) \\ \phi \equiv 0 \quad (e-g)/2 - if &\equiv 0 \quad e = g \quad f = 0 \quad 3-2 \quad 4 S \quad \text{Hopf} \end{aligned}$$

### 1.3 Complete Surfaces. Theorem of Hopf-Rinow

5-2  $[ \quad ]$   $S$  extendable  $\bar{S} \subset S$   $\bar{S} \subset S$  nonextendable

$R$   $[ \quad ]$   $S$  complete  $p \in S$   $p = \gamma(0)$   $\gamma : [0, \epsilon) \rightarrow S$   
 $\tilde{\gamma} : R \rightarrow S$   
 $p \in S$   $\exp_p : T_p(S) \rightarrow S$   $v \in T_p(S)$

$\cdot$   
 $\cdot \Rightarrow \Rightarrow 5$   
 $S$

$\bar{S} \neq S$   $S$   $S$   $\bar{S}$   $\text{Bd } S$   $p \in \text{Bd } S$   $p \notin S$   $\bar{S}$   $p$   
 $\bar{V}$   $p$   $p \in \text{Bd } S$   $q_0 \in \bar{V} \cap S$   $\bar{\gamma}$   $p$   $q_0$   $\alpha(t) = \bar{\gamma}(1-t)$   $S$   $q_0$   
 $t=1$   $p \notin S$   $^4$   
 $p_0$   $S$   $p_0$   $p_0$  ———  $\text{fig:fig5-}$

5  $\text{fig:fig5-6}$

$[ \quad ]$   $S$   $p, q \in S$   $\alpha_{p,q}$   $p$   $q$   $l(\alpha_{p,q})$   $p$   $q$   
 $d(p, q) = \inf_{\alpha_{p,q}} l(\alpha_{p,q}),$

1.  $d(p, q) = d(q, p)$
2.  $d(p, q) + d(q, r) \geq d(p, r)$
3.  $d(p, q) \geq 0$   $d(p, q) = 0$   $p = q$

$S \subset R^3$   
 $\gamma : [0, \epsilon) \rightarrow S$   $p$   $R$   $\bar{\gamma}$   $s < s_0$   $s_0$   $\{s_n\}$   
 $\{\bar{\gamma}(s_n)\}$   $q \in S$   $\bar{\gamma}$   $s_0$  connectedness of  $R$   $^5$

$\cdot$   
 $\text{fig:fig5-7}$   
 $p, q \in S$  minimal geodesic  $l(\gamma) = d(p, q)$   
 $[\text{Hopf-Rinow}]$   $S$   $p, q \in S$   $p$   $q$   
 $[ \quad ]$   $r = d(p, q)$   $T_p(S)$   $\delta$   $\exp_p$   $\Sigma = \text{Bd } B_\delta(p)$   $\Sigma$   $d(x, q)$   
 $x_0$   $p$   $v = x_0 - p$   $\gamma(s) = \exp_p(sv)$   $\gamma(r) \neq q$   
 $\gamma(r) = q$   $l(\gamma) = r = d(p, q)$   $^6$   
 $S$   $\exp_p : T_p(S) \rightarrow S$

$S^2$   $\rho(S^2) = \pi$   
 $S - \{p\}$   $p$   $p$   $\text{fig:fig5-3}$   
Hopf-Rinow

<sup>4</sup> sec:derivation-complete-nonextendable

<sup>5</sup> sec:derivation-closed-complete

<sup>6</sup> sec:derivation-hopf-rinow

[Figure 5-2.

][width=]../../PDFs/differential-  
 geometry/transcript/Do Carmo -  
 Differential Geometry of Curves and  
 Surfaces/hybrid<sub>o</sub>cr/images/d0706b198f2897c57beca8da2254a5745b2ff1562571473ec50989e16a

[Figure 5-1a.  $P$  " "  $S'$   
 S']][width=]../../PDFs/differential-  
 geometry/transcript/Do Carmo -  
 Differential Geometry of Curves  
 and

Surfaces/hybrid<sub>o</sub>cr/images/1010f8a78b7253968f731675b5aa9f43bf3ca471ea4bd8f01af07ba38d735e75.jpg  
 Surfaces/hybrid<sub>o</sub>cr/images/c2b845abcc71c94b3bb04390b4cc2df940897

[Figure 5-1b. " "  $S''$   $S'$   $S''$   
 $S''$ 

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 geometry/transcript/Do Carmo -  
 Differential Geometry of Curves

[Figure 5-5.  $p$   $p$ 

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 geometry/transcript/Do Carmo -  
 Differential Geometry of Curves  
 and

Surfaces/hybrid<sub>o</sub>cr/images/beb5e685031a4887db06d1c883d81f99a5afeb21221367a9f0f8e0

[Figure 5-7.

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 Differential Geometry of Curves and  
 Surfaces/hybrid<sub>o</sub>cr/images/acad36bb78b1fb869d9d94c12cba6f4b336f5a326660d438ba1de2cd31

[Figure 5-9. Hopf-Rinow

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 Differential Geometry of Curves and  
 Surfaces/hybrid<sub>o</sub>cr/images/b76a37d9c31c41a342040fb614134337cbc63dd6b8139d1993a13b3af5

[Figure 5-3.  $S - \{p\}$ 

geodesic  $\gamma$   $p$   $p \notin S -$   
 $\{p\}$ ][width=]../../PDFs/differential-  
 geometry/transcript/Do  
 Carmo - Differential Geometry  
 of Curves and

Surfaces/hybrid<sub>o</sub>cr/images/d7bfebe261ddf7bf614425e1e3b7a806239728d58b983dc44364b

## 5-3

5-3, 1 — do Carmo, Exercise 5-3, 1  $S \subset \mathbb{R}^3$   $F \subset S$   $S - F$   $S - F$   
 $: F \subset S$  " "  $F \subset S - F$   $S - F$   $p \in F$   $p \in S - F$   
 $S \subset \mathbb{R}^3$   $p \in S$   $p \in F$   $p \in S - F$   $p \in S$   $p \in F$   $p \in S - F$   
 $: S \subset \mathbb{R}^3$   $F \subset S$   $S - F$   $p \in F$   $F \subset S - F$   $p \in S - F$   $S \subset \mathbb{R}^3$   $p \in S$   
 $R \subset \mathbb{R}^3$   $p \in R$   $\gamma: [0, \infty) \rightarrow S$   $\gamma \cap F = \emptyset$   $\gamma \cap (S - F) = \gamma$   $S - F$   
 $F \subset \mathbb{R}^3 - F$   $q \in F$  Hopf-Rinow  $\gamma \cap F = \emptyset$   $\gamma \cap (S - F) = \gamma$   $q \in F$   $r \notin F$   $\gamma \cap q$   
 $F \subset S - F$  do Carmo  $F \neq \emptyset$   $p \in F$   $p \in S - F$   
 $\Sigma \subset S - F$   $p \in \Sigma$   $p \in S - F$   
 5-3, 2 — do Carmo, Exercise 5-3, 2  $S$  ??  $p \in S$   $p \in S$   
 $p \in S$   $: S = \{z = \sqrt{x^2 + y^2}\}$   $p_0 \in S$   $p_0 \in S$   
 Clairaut  $r \cos \alpha = \text{const}$   
 $: z = \sqrt{x^2 + y^2}$

$$\mathbf{x}(u, v) = (v \cos u, v \sin u, v), \quad u \in (0, 2\pi), v > 0.$$

3-3 Clairaut  $v \cos \alpha = \text{const}$   $\alpha$   $v$

- $\text{const} = 0$   $\cos \alpha = 0$   $\alpha = \pi/2$   $u = \text{const}$   $v = s$   $s \in (0, \infty)$   
 $v \rightarrow 0^+$   $p_0 \in S$   $v = 0$
- $\text{const} > 0$   $\cos \alpha > 0$   $\alpha < \pi/2$   $v \rightarrow \infty$   
 $v \rightarrow 0^+$  Clairaut  $v \rightarrow 0^+$   $\cos \alpha \rightarrow \text{const}/0$   $\text{const} = 0$   $\alpha$   
 $\pi/2$   $\text{const}$   $\text{const} > 0$   $p_0$

5-3, 3 — do Carmo, Exercise 5-3, 3  $S$  ?? 4-2 3  $p, q \in S$   $S$   
 fig:fig5-11  $: S$  4-2 3  $S \setminus \gamma$   $W = \{(r, \theta) : 0 < r < \infty, \alpha_0 < \theta < \alpha_1\}$   $\alpha_1 - \alpha_0 = 2\pi \sin \beta < 2\pi$   $\varphi: S \setminus \gamma \rightarrow W$   $p, q \in S$   
 (1)  $p, q \in \gamma$   $\varphi(p), \varphi(q) \in W$   $W$   $\overline{\varphi(p)\varphi(q)} \subset W$   $\varphi^{-1}(\overline{\varphi(p)\varphi(q)})$   
 $p, q$   
 (2)  $p \in \gamma$   $q \in \gamma$   $p \in \gamma$   $q \in \gamma$   $S \setminus \gamma_1$   $\gamma_1$   $p \in \gamma_1$   
 $p, q$

5-3, 4 — do Carmo, Exercise 5-3, 4  $S$   $\{p_n\}$   $d$   $p_0 \in S$   $\epsilon > 0$   
 $n_0$   $n \geq n_0$   $d(p_n, p_0) < \epsilon$   $S$   $\{p_n\}$   $d$   $p_0$   $\{p_n\}$   $\mathbb{R}^3$   $p_0$   $:$   
 $d$   $\|p - q\|$   $d(p, q) \geq \|p - q\|$   
 $: \{p_n\}$   $d$   $p_0$

$$\|p_n - p_0\| \leq d(p_n, p_0) \rightarrow 0,$$

$\{p_n\}$   $p_0$   $p_0$   $V$   $\exp_{p_0} B_\delta(0) \subset T_{p_0}(S)$   $n_0$   $n \geq$   
 $n_0$   $p_n \in V$   $d(p_n, p_0) = |\exp_{p_0}^{-1}(p_n)|$   $V$   $C > 0$   
 $d(p_n, p_0) \leq C\|p_n - p_0\|$   $d(p_n, p_0) \rightarrow 0$   $\{p_n\}$   $d$



5-3, 5\* — do Carmo, Exercise 5-3, 5  $S$   $S$  Cauchy  $S$  :  
 Cauchy " "  
 :  $S$   $S$  Cauchy  $\{p_n\}$   $\{p_n\}$   $p \in S$   
 $r_n = d(p_1, p_n)$   $v_n \in T_{p_1}(S)$   $p_n = \exp_{p_1}(r_n v_n)$   $|v_n| = 1$   $\{p_n\}$  Cauchy  
 $n_0$   $n, m \geq n_0$   $d(p_n, p_m) < \delta$   $B_\delta(p_1)$   $d(p_1, \cdot) = |\exp_{p_1}^{-1}(\cdot)|$   
 $\|r_n v_n - r_m v_m\| \leq d(p_n, p_m) < \delta,$   
 $\{r_n v_n\}$   $T_{p_1}(S)$  Cauchy  $T_{p_1}(S) \cong R^2$   $v \in T_{p_1}(S)$   $r_n v_n \rightarrow v$   
 $r = |v|$   $q = \exp_{p_1}(v)$   $p_n \rightarrow q$  Cauchy  
 $S$  Cauchy  $S$   $p$   $\gamma : [0, \epsilon) \rightarrow S$   $s = 1$   $l = 1$   
 $t_n \rightarrow 1$   $t_n < 1$   $p_n = \gamma(t_n)$   $\{p_n\}$  Cauchy  $\gamma$   $d(p_n, p_m) = |t_n - t_m| \rightarrow$   
 $0$   $q \in S$   $\gamma$   $p_0 = \lim \gamma(t_n)$   $\gamma$   $t = 1$   
 5-3, 6\* — do Carmo, Exercise 5-3, 6  $S$   $p$  ray  $l(\gamma) = d(\gamma(0), \gamma(s))$   
 $s \geq 0$   $\gamma : [0, \infty) \rightarrow S$   $S$   $S$   $p$  : " " " " —  
 —  
 :  $S$   $p$   $p$   $v \in T_p(S)$   $|v| = 1$   $\gamma_v$   $v$   $v$   $\gamma_v$   
 $d(p, \gamma_v(s)) = s$   $s \geq 0$   $v$   $s_v > 0$   $d(p, \gamma_v(s_v)) < s_v$   
 $T_p(S)$   $S^1$   $s_v$   $s_0 > 0$   $v$   $s_v$   $\gamma_v$   $[0, s_v]$   $s_v$   $p$   
 $\gamma_v$   $v$  do Carmo  $d(p, \cdot)$   $q_n \in S$   $d(p, q_n) \rightarrow \infty$   
 $p$   $q_n$   $\gamma_n$  Hopf-Rinow  $l_n = d(p, q_n)$   $v_n \in T_p(S)$   $S^1$   
 $v_{n_k} \rightarrow v$   $\gamma_{n_k} \rightarrow \gamma_v$   $\gamma_v$   $v$   $\gamma_v$   $s_0$   $d(p, \gamma_v(s_0)) < s_0$   
 $n_k$   $l_{n_k} > s_0$   $\gamma_{n_k}$   $\gamma_v$   
 5-3, 7 — do Carmo, Exercise 5-3, 7  $S \subset R^3$  divergent curve  
 :  
 :  $S$   $\alpha : [0, \infty) \rightarrow S$   $|\alpha'|$   $\alpha$   $\alpha$   $S$   
 $p$   $\gamma : [0, \epsilon) \rightarrow S$   $l(\gamma) = \epsilon$   $s = \epsilon + \delta$   $\delta_n \rightarrow 0$   $p_n = \gamma(\epsilon - \delta_n)$   
 $p_n \rightarrow p_\epsilon = \lim_{s \rightarrow \epsilon} \gamma(s) \in R^3$   $p_\epsilon \notin S$   $\beta_n$   $p_n$   $\gamma$   $p_\epsilon$   $\beta_n$   
 do Carmo  $S$   $p \in S$   $\exp_p$   $v \in T_p(S)$   $|v| = r$   
 $v_n \rightarrow v$   $v_n \in T_p(S)$   $|v_n| < r$   $p_n = \exp_p(v_n) \in S$   $p_n \rightarrow q = \lim \exp_p(v_n)$   $q \in$   
 $R^3$   $q \notin S$   $p_n$   $q$   $S$   $\alpha$   $\alpha$   $S$  " "  $\alpha$   
 8  
 5-3, 8\* — do Carmo, Exercise 5-3, 8  $S$   $\bar{S}$   $\varphi : S \rightarrow \bar{S}$   $\bar{S}$   
 $c > 0$   $I_p(v) \geq c \bar{I}_{\varphi(p)}(d\varphi_p(v))$   $p \in S$   $v \in T_p(S)$   $S$  :  
 $I_p(v) \geq c \bar{I}_{\varphi(p)}(d\varphi_p(v))$   $\varphi$   $\bar{S}$   $\varphi(p)$   $S$   $\bar{S}$   
 :  $I_p(v) \geq c \bar{I}_{\varphi(p)}(d\varphi_p(v))$   $p$   $v$

$$|d\varphi_p(v)|_{\bar{S}} \leq \frac{1}{\sqrt{c}} |v|_S.$$

$$\gamma : [0, a) \rightarrow S \quad \bar{\gamma} = \varphi \circ \gamma \quad \bar{S}$$

$$|\bar{\gamma}'(s)|_{\bar{S}} \leq \frac{1}{\sqrt{c}} |\gamma'(s)|_S = \frac{1}{\sqrt{c}}.$$

$$\begin{aligned}
& \bar{\gamma} \quad \bar{l}(s) = \int_0^s |\bar{\gamma}'| ds \leq \frac{s}{\sqrt{c}} \quad \bar{S} \quad \bar{\gamma} \quad R \quad \gamma \quad R \quad S \quad \varphi \\
& I \quad \varphi \quad d\varphi \quad \gamma \quad S \\
& 5-3, 9^* \text{ — do Carmo, Exercise 5-3, 9} \quad S_1 \subset R^3 \quad S_2 \subset R^3 \\
& \varphi : S_1 \rightarrow S_2 \quad \varphi \quad : S_1 \quad \text{Cauchy} \quad S_2 \quad \text{Hadamard} \\
& S_2 \cong R^2 \quad \varphi \quad \varphi \quad S_1 \quad S_2 \quad S_2 \quad \text{Hadamard} \quad \varphi \quad d_1 \\
& \quad d_2 \\
& : \quad \text{Hadamard} \quad 5-6 \quad 2 \quad S_2 \quad K \leq 0 \quad \exp_q : T_q(S_2) \rightarrow S_2 \quad q \in S_2 \\
& \quad S_2 \quad \exp_q \quad \exp_q \quad S_2 \quad R^2
\end{aligned}$$

$$\begin{aligned}
& l \quad S_1 \quad p \in S_1 \quad q = \varphi(p) \quad S_1 \\
& \alpha \quad |\varphi \circ \alpha'(t)|_{S_2} = |\alpha'(t)|_{S_1} \quad L_{S_2}(\varphi \circ \alpha) = L_{S_1}(\alpha) \quad \varphi \quad d_2(\varphi(p), \varphi(q)) = \\
& d_1(p, q) \\
& \quad \varphi \quad r \in S_2 \quad S_2 \cong R^2 \quad q \quad r \quad \gamma \quad \tilde{\gamma} \quad T_p(S_1) \quad d \exp_p(\tilde{\gamma}(0)) = \\
& \gamma'(0) \quad \varphi \quad \varphi(\exp_p(\tilde{\gamma}(t))) = \exp_q(\tilde{\gamma}(t)) \quad \tilde{\gamma} \quad \varphi \quad S_2 \\
& \quad d_2(\varphi(p_1), \varphi(p_2)) = d_1(p_1, p_2) \quad p_1, p_2 \in S_1 \quad \varphi \\
& 5-3, 10^* \text{ — do Carmo, Exercise 5-3, 10} \quad S \subset R^3 \quad v \quad h(p) = \langle p, v \rangle \\
& \alpha(t) \quad \text{grad } h \quad a \quad |\text{grad } h(p)| \leq 1 \quad b \quad \alpha(t) \quad t \in R \quad : \quad h(p) = \langle p, v \rangle \\
& \quad \text{grad } h \quad h \quad S \subset R^3 \quad h \quad S \quad 1 \quad |p| \geq h(p) \quad \alpha \quad \alpha
\end{aligned}$$

$$\begin{aligned}
& : (a) \quad \langle \text{grad } h(p), w \rangle = dh_p(w) = \langle p, w \rangle \quad w \in T_p(S) \quad dh_p(w) = \\
& d\langle p, v \rangle(w) = \langle w, v \rangle \quad T_p(S) \quad w \quad \text{grad } h(p) \quad v \quad v \quad |dh_p| = \\
& |v|^T \leq |v| = 1 \quad |\text{grad } h(p)| \leq 1 \quad v \quad N(p) \\
& (b) \quad \alpha(t) \quad \text{grad } h \quad \alpha'(t) = \text{grad } h(\alpha(t)) \quad (a) \quad |\alpha'(t)| \leq 1
\end{aligned}$$

$$|\alpha(t) - \alpha(0)| \leq \int_0^t |\alpha'(s)| ds \leq t.$$

$$\begin{aligned}
& S \quad \alpha \quad |\alpha'(t)| \leq 1 \quad \alpha \quad R \quad h \quad R^3 \quad \text{grad } h \\
& \text{Lipschitz} \quad \text{Lipschitz} \quad 1 \quad \text{ODE} \quad \alpha(t) \quad t \in R \quad S \quad \text{grad } h \quad S \\
& \alpha(t) \in S \quad t \\
& 5-3, 11^* \text{ — Osserman} \quad , \text{ do Carmo, Exercise 5-3, 11} \quad \mathbf{x} : D_1 \rightarrow R^3 \\
& \quad H = 0 \quad \epsilon > 0 \quad w \quad |\langle \mathbf{x}_u, w \rangle| / |\mathbf{x}_u| \geq \epsilon \quad |\langle \mathbf{x}_v, w \rangle| / |\mathbf{x}_v| \geq \\
& \epsilon \quad \mathbf{x}(D_1) \quad \text{Osserman} \quad 3-5 \quad : \text{Osserman} \\
& \theta \quad \theta^{-1} \\
& : (a) \quad \mathbf{x} : D_1 \rightarrow R^3 \quad E = G = \lambda^2 \quad F = 0 \quad \mathbf{x}_{uu} + \mathbf{x}_{vv} = 0 \\
& \quad H = 0 \quad \varphi(\zeta) = \langle \mathbf{x}_u, w \rangle + i \langle \mathbf{x}_v, w \rangle
\end{aligned}$$

$$\varphi_{\bar{\zeta}} = \frac{1}{4} [(\mathbf{x}_{uu} + \mathbf{x}_{vv}) \cdot w - i(\mathbf{x}_{uu} + \mathbf{x}_{vv}) \cdot w] = 0,$$

$$\begin{aligned}
& \varphi \quad D_1 \\
& (b) \quad \theta(\zeta) = \int_0^\zeta \varphi(\zeta) d\zeta \quad (a) \quad \theta \quad \theta(0) = 0 \quad (*) \quad |\varphi(\zeta)| \geq \epsilon |\mathbf{x}_u| \geq \epsilon \lambda \\
& |\mathbf{x}_u| = |\mathbf{x}_v| = \lambda \quad |\langle \mathbf{x}_u, w \rangle| \geq \epsilon |\mathbf{x}_u| \quad \theta'(\zeta) = \varphi(\zeta) \neq 0 \quad \theta \quad \theta^{-1} \quad \theta^{-1} \\
& C \quad \theta^{-1} \quad |\theta^{-1}(\eta)| \leq R \quad \text{Liouville} \quad \theta^{-1} \quad \theta^{-1} \quad \eta_0 \quad |\eta_0| = R \\
& (c) \quad L = \{t\eta_0 : 0 \leq t \leq 1\} \quad \alpha = \theta^{-1}(L)
\end{aligned}$$

$$|\mathbf{x}_s| = \lambda \sqrt{u_s^2 + v_s^2}, \quad d\zeta^2 = (du)^2 + (dv)^2.$$

#### 1.4. FIRST AND SECOND VARIATIONS OF ARC LENGTH; BONNET'S THEOREM

$$\begin{aligned}
 |\mathbf{x}_u| &= |\mathbf{x}_v| = \lambda \quad |\mathbf{x}_s|^2 = \lambda^2((u_s)^2 + (v_s)^2) \quad (*) \\
 |\langle \mathbf{x}_u, w \rangle|^2 + |\langle \mathbf{x}_v, w \rangle|^2 &\geq \epsilon^2 |\mathbf{x}_u|^2 = \epsilon^2 \lambda^2. \\
 |\varphi|^2 &= |\langle \mathbf{x}_u, w \rangle|^2 + |\langle \mathbf{x}_v, w \rangle|^2 \geq \epsilon^2 \lambda^2 \\
 l &= \int_{\alpha} |\mathbf{x}_s| ds = \int_{\alpha} \lambda \sqrt{u_s^2 + v_s^2} |d\zeta| = \int_{\alpha} \sqrt{\langle \mathbf{x}_u, \mathbf{x}_u \rangle u_s^2 + \cdots} |d\zeta| \leq \frac{1}{\epsilon} \int_{\alpha} |\varphi| |d\zeta| = \frac{1}{\epsilon} \int_0^R |\varphi(\theta^{-1}(\eta))| |d\eta| = \frac{R}{\epsilon} < \infty \\
 \Rightarrow \quad \mathbf{x}(\alpha) & \text{ Exercise 7 } \mathbf{x}(D)
 \end{aligned}$$

#### 1.4 First and Second Variations of Arc Length; Bonnet's Theorem

$$\begin{aligned}
 \cdot \quad \text{Bonnet} \quad S \quad K &\geq \delta > 0 \quad S \\
 \cdot \quad K &\geq \delta > 0 \quad \pi/\sqrt{\delta} \\
 \cdot \quad \alpha : [0, l] &\rightarrow S \quad \alpha \quad \text{variation} \quad h : [0, l] \times (-\epsilon, \epsilon) \rightarrow S \\
 h(s, 0) &= \alpha(s) \quad h(0, t) = \alpha(0) \quad h(l, t) = \alpha(l) \quad t \quad \text{proper variation} \quad \text{variational} \\
 \text{vector field} \quad V(s) &= (\partial h / \partial t)(s, 0) \\
 L(t) &= \int_0^l |\partial h / \partial s(s, t)| ds \\
 L(t) \quad t &= 0 \\
 L'(0) &= - \int_0^l \langle A(s), V(s) \rangle ds, \\
 A(s) &= (D/\partial s)(\partial h / \partial s)(s, 0) \quad \alpha \\
 L'(0) \quad V(s) \quad A(s) \quad \alpha \\
 \alpha \quad L'(0) &= 0 \\
 \cdot \quad \text{---} \\
 \langle V(s), \gamma'(s) \rangle &= 0 \\
 h : [0, l] \times (-\epsilon, \epsilon) &\rightarrow S \quad V(s, t) \quad h \\
 \frac{D}{\partial t} \frac{D}{\partial s} V - \frac{D}{\partial s} \frac{D}{\partial t} V &= K(s, t) \left( \frac{\partial h}{\partial s} \wedge \frac{\partial h}{\partial t} \right) \wedge V, \\
 K(s, t) \quad S \quad h(s, t) \\
 h \quad \gamma : [0, l] &\rightarrow S \quad V(s) = (\partial h / \partial t)(s, 0) \\
 L''(0) &= \int_0^l \left( \left| \frac{D}{\partial s} V(s) \right|^2 - K(s) |V(s)|^2 \right) ds, \\
 K(s) &= K(s, 0) \\
 \begin{bmatrix} & \\ & \end{bmatrix} \quad L''(0) \quad \gamma \quad A(s) &= 0 \quad \langle \partial h / \partial s, \partial h / \partial t \rangle = 0 \quad ?? \quad 9 \\
 4 \quad V(s) &
 \end{aligned}$$

$$L''(0) = - \int_0^l \left\langle \frac{D^2 V}{ds^2} + KV, V \right\rangle ds.$$

Bonnet  
[Bonnet]  $S \quad K \geq \delta > 0 \quad S \quad \rho$

$$\rho \leq \frac{\pi}{\sqrt{\delta}}.$$

$\begin{bmatrix} \cdot \\ \cdot \end{bmatrix} \quad p, q \in S \quad \gamma : [0, l] \rightarrow S \quad \text{Hopf-Rinow} \quad l > \pi/\sqrt{\delta}$   
 $w(s) \quad \gamma$

$$V(s) = w(s) \sin \frac{\pi}{l} s, \quad s \in (0, l].$$

$$V(0) = V(l) = 0 \quad \langle V(s), \gamma'(s) \rangle = 0 \quad ??$$

$$L''(0) = \int_0^l \left( \frac{\pi^2}{l^2} \cos^2 \frac{\pi}{l} s - K(s) \sin^2 \frac{\pi}{l} s \right) ds.$$

$$\frac{K \geq \delta > \pi^2/l^2}{\pi/\sqrt{\delta}} \quad ?? \quad S \quad 10 \quad \gamma \quad L''(0) \geq 0 \quad l \leq \pi/\sqrt{\delta} \quad S \leq$$

Bonnet  $K \equiv 1 \quad \rho = \pi$   
 $K \geq \delta > 0 \quad K > 0 \quad z = x^2 + y^2 \quad K > 0 \quad \text{---} \quad K$   
 5-5, 1 — do Carmo, Exercise 5-5, 1 a  $\gamma : [0, l] \rightarrow S \quad J(s) \quad \gamma \quad \text{Jacobi}$   
 $J(0) = 0 \quad \langle J'(0), \gamma'(0) \rangle = 0 \quad \langle J(s), \gamma'(s) \rangle = 0 \quad s \quad \text{b} \quad |J'(0)| = 1$   
 $\{e_1(s), e_2(s)\} \quad \gamma \quad e_1(0) = \gamma'(0) \quad e_2(0) = J'(0) \quad J(s) = u(s)e_2(s) \quad u(s)$   
 $u'' + K(s)u = 0 \quad u(0) = 0 \quad u'(0) = 1 \quad : \quad \text{(a)} \quad \text{Jacobi} \quad \text{---}$   
 Jacobi  $\langle DJ/ds, \gamma' \rangle \quad \text{(b)} \quad \text{Jacobi} \quad \text{ODE}$   
 $: \quad \text{(a)} \quad \gamma \quad \text{Jacobi} \quad \frac{D^2 J}{ds^2} + KJ = 0 \quad K \quad \langle J(s), \gamma'(s) \rangle$

$$\frac{d}{ds} \langle J, \gamma' \rangle = \langle \frac{DJ}{ds}, \gamma' \rangle + \langle J, \frac{D\gamma'}{ds} \rangle = \langle \frac{DJ}{ds}, \gamma' \rangle,$$

$$\gamma \quad \frac{D\gamma'}{ds} = 0$$

$$\frac{d^2}{ds^2} \langle J, \gamma' \rangle = \langle \frac{D^2 J}{ds^2}, \gamma' \rangle = -K \langle J, \gamma' \rangle.$$

$$f(s) = \langle J(s), \gamma'(s) \rangle \quad f'' + K(s)f = 0 \quad f(0) = 0 \quad J(0) = 0$$

$$f'(0) = \langle J'(0), \gamma'(0) \rangle = 0$$

ODE  $f(s) \equiv 0 \quad \langle J(s), \gamma'(s) \rangle = 0$   
 (b)  $\{e_1(s), e_2(s)\} \quad \gamma \quad e_1(0) = \gamma'(0) \quad e_2(0) = J'(0) \quad J(s) \quad J(s) =$   
 $a(s)e_1(s) + b(s)e_2(s) \quad \text{(a)} \quad \langle J, \gamma' \rangle = 0 \quad e_1(s) = \gamma'(s) \quad a(s) \equiv 0 \quad J(s) =$   
 $b(s)e_2(s) \quad \text{Jacobi} \quad e_2(s) \quad b'' + K(s)b = 0 \quad J(0) = 0 \quad b(0) = 0$   
 $|J'(0)| = 1 \quad e_2(0) = J'(0) \quad b'(0) = 1$   
 5-5, 2 — do Carmo, Exercise 5-5, 2  $z = x^2 + y^2 \quad p = (0, 0, 0)$   
 $: \quad z = x^2 + y^2 \quad K > 0 \quad \text{Jacobi} \quad \text{---} \quad (K = 0)$

1.4. FIRST AND SECOND VARIATIONS OF ARC LENGTH; BONNET'S THEOREM 13

$$z = x^2 + y^2 \quad E = G = 1 + 4(x^2 + y^2) \quad F = 0$$

$$K = \frac{4}{(1 + 4r^2)^2}, \quad r^2 = x^2 + y^2.$$

Jacobi  $u'' + K(s)u = 0$  1  $K(s) > 0$   $K(s) \rightarrow 0$   $s \rightarrow \infty$   
 $u(s)$   $u'' \leq 0$   $K$   $u(s)$   $v(s)$   $v'' = 0$   $v(0) =$   
 $0$   $v'(0) = 1$   $v(s) = s$   $K > 0$  3  $u(s) \geq v(s) = s > 0$   $u$   $p$   
 5-5, 3\* — do Carmo, Exercise 5-5, 3  $S$   $\tilde{S}$   $p \in S$   $\tilde{p} \in \tilde{S}$   $i : T_p(S) \rightarrow$   
 $T_{\tilde{p}}(\tilde{S})$   $\gamma : [0, \infty) \rightarrow S$   $J(s)$   $\gamma$  Jacobi  $J(0) = 0$   $\langle J'(0), \gamma'(0) \rangle =$   
 $0$   $|J'(0)| = 1$  a  $J(s) = v(s)e_2(s)$   $\tilde{J}(s) = u(s)\tilde{e}_2(s)$   $v$   $u$   $v'' + K(s)v =$   
 $0$   $v(0) = 0$   $v'(0) = 1$   $u'' + \tilde{K}(s)u = 0$   $u(0) = 0$   $u'(0) = 1$  b  $K(s) \leq$   
 $\tilde{K}(s)$   $s \in [0, \infty)$   $\tilde{\gamma}$   $s = a$   $\gamma$   $s = a$  c  $|J(s)| \geq |\tilde{J}(s)|$   
 $s \in [0, a)$  d (c)  $K(s) = \tilde{K}(s)$  : Jacobi  
 Jacobi " " Jacobi Sturm  
 : (a) 1  $\langle J(s), \gamma'(s) \rangle \equiv 0$   $J(s) = v(s)e_2(s)$   $v$   $v'' + K(s)v = 0$   $v(0) =$   
 $0$   $v'(0) = |J'(0)| = 1$   $\tilde{J}(s) = u(s)\tilde{e}_2(s)$   
 (b) a  $\tilde{J}$   $s_0 \in (0, a)$   $\tilde{v}(s) = u(s)$   $v(s)$   $s \in [0, s_0]$   $u(s) > 0$   
 $(0, a)$   $K \leq \tilde{K}$  Sturm 6  $v(s)$   $[0, s_0]$   $s_0 < a$   $\gamma$   $a$   
 (c) Gauss  $|J(s)| = |v(s)|$   $|\tilde{J}(s)| = |u(s)|$  (b)  $K \leq \tilde{K}$   $v(s) \geq$   
 $u(s)$   $|J(s)| \geq |\tilde{J}(s)|$   
 (d)  $K(s) = \tilde{K}(s)$   $v$   $u$   $v \equiv u$   $|J(s)| = |\tilde{J}(s)|$   $s$   
 $v \equiv u$   $K \equiv \tilde{K}$   
 5-5, 4 — do Carmo, Exercise 5-5, 4  $S$   $K \leq K_0$   $K_0 > 0$   
 $\gamma : [0, \infty) \rightarrow S$   $(0, \pi/\sqrt{K_0}]$  :  $K = K_0$   $s = \pi/\sqrt{K_0}$   
 $S$   $K \leq K_0$   
 :  $\tilde{S}$   $K_0$  3(b)  $\tilde{S}$   $s = \pi/\sqrt{K_0}$   $S$   
 $K_0$   $u'' + K_0u = 0$   $u(0) = 0$   $u'(0) = 1$   $u(s) = \frac{1}{\sqrt{K_0}} \sin(\sqrt{K_0}s)$   $s =$   
 $\pi/\sqrt{K_0}$   $S$   $(0, \pi/\sqrt{K_0})$   
 5-5, 5 — do Carmo, Exercise 5-5, 5  $S$   $K \geq K_1 > 0$  " "  $\gamma :$   
 $[0, \infty) \rightarrow S$   $(0, \pi/\sqrt{K_1}]$  : 4  $K_1$  " "  
 :  $u$   $u'' + K_1u = 0$   $u(0) = 0$   $u'(0) = 1$   $u(s) = \frac{1}{\sqrt{K_1}} \sin(\sqrt{K_1}s)$   
 $s = \pi/\sqrt{K_1}$   $K(s) \geq K_1$  3(c)  $J$   $|J(s)| \leq u(s)$   $J$   
 $s = \pi/\sqrt{K_1}$   $(0, \pi/\sqrt{K_1}]$   
 5-5, 6\* Sturm — do Carmo, Exercise 5-5, 6  $S$   $\gamma : [0, \infty) \rightarrow S$   $J(s)$   
 $\gamma$  Jacobi  $J(0) = J(s_0) = 0$   $s_0 \in (0, \infty)$   $J(s) \neq 0$   $s \in (0, s_0)$   $K(s) \leq$   
 $L(s)$   $L$   $[0, \infty)$   $u'' + L(s)u = 0$   $(0, s_0]$  : Sturm  
 ODE  $u$   $u'' + Lu = 0$   $v$   $v'' + Kv = 0$   $K \leq L$   $v$   $s_0$   $u$   $s_0$   
 $J$   $v$   
 :  $v(s) = |J(s)|$  Jacobi  $K \leq L$   $v'' + Lv \leq 0$   $v'' + Kv = 0$   
 $v'' + Lv = -L_v + Lv \leq Kv + Lv \leq 0$   $v > 0$   $u$   $u'' + Lu = 0$   $w = u/v$   
 $s = 0$   $s = s_0$   $w$   $(vw')' = w(v'' + Lu) \leq 0$   $(vw')$   $[0, s_0]$   
 $v(0) = 0$   $v(s_0) = 0$   $(vw')$   $[0, s_0]$

$$0 = (vw')|_{s_0} - (vw')|_0 = - \int_0^{s_0} (vw')' ds \geq 0.$$

$u(0) = 0$   $u$   $s_0$   $w$   $(vw')$   $u$   $(0, s_0)$   
 5-5, 7\* Kneser — do Carmo, Exercise 5-5, 7  $\gamma : [0, \infty) \rightarrow S$   $S$   
 $\gamma(0) = p$   $K(s)$   $\gamma$   $\int_t^\infty K(s) ds \leq \frac{1}{4(t+1)}$   $t \geq 0$  a  $w(t) =$   
 $\int_t^\infty K(s) ds + \frac{1}{4(t+1)}$   $w'(t) + w(t)^2 \leq -K(t)$  b  $w'(t) + w(t)^2 = -L(t)$   $L(t) \geq$   
 $K(t)$   $v(t) = \exp(\int_0^t w(s) ds)$   $v''(t) + L(t)v(t) = 0$   $v(0) = 1$   $v'(0) = 0$  c  
 Sturm  $\gamma$   $p$  : Kneser  $w$  — Sturm

: (a)

$$w'(t) = -K(t) - \frac{1}{4(t+1)^2}, \quad w(t)^2 = \left( \int_t^\infty K + \frac{1}{4(t+1)} \right)^2 \geq \frac{1}{4(t+1)^2},$$

$$w'(t) + w(t)^2 \leq -K(t)$$

(b)  $v(t)$

$$v'(t) = w(t)v(t), \quad v''(t) = (w'(t) + w(t)^2)v(t) = -L(t)v(t).$$

$$v(0) = \exp(0) = 1 \quad v'(0) = w(0) \cdot 1 = 0$$

(c) (b)  $v$   $v'' + Lv = 0$   $v(0) = 1$   $v'(0) = 0$   $L(t) \geq K(t)$   $u'' + Ku = 0$   
 Jacobi Sturm  $v$   $K$  Jacobi  $\gamma$   $p$   
 5-5, 8\* — do Carmo, Exercise 5-5, 8  $\gamma : [0, l] \rightarrow S$   $S$   $\gamma(l)$   $\gamma(0)$   
 $w_0 \in T_{\gamma(0)}(S)$   $w_1 \in T_{\gamma(l)}(S)$   $\gamma$  Jacobi  $J(s)$   $J(0) = w_0$   $J(l) = w_1$  :  
 Jacobi ODE 4  $(J(0), J'(0))$   $J(0) = w_0$   $J(l) = w_1$  —  
 $\gamma(l)$   $\gamma(0)$   $(J(0), J'(0)) \mapsto (J(0), J(l))$   
 : Jacobi  $\frac{D^2 J}{ds^2} + KJ = 0$   $\gamma$   $\{e_1(s), e_2(s)\}$   $e_1(0) = \gamma'(0)$   
 $J(s) = a(s)e_1(s) + b(s)e_2(s)$  b  $b'' + K(s)b = 0$  a  $a'' + K(s)a = 0$   
 $a(0), a(l), b(0), b(l)$   $w_0 = w_1 = 0$  — Jacobi  
 $\gamma(l)$   $\gamma(0)$

5-5, 9 — do Carmo, Exercise 5-5, 9  $J(s)$   $\gamma : [0, l] \rightarrow S$  Jacobi  
 $\langle J(0), \gamma'(0) \rangle = 0$   $J'(0) = 0$   $\langle J(s), \gamma'(s) \rangle = 0$  s : 1(a)  
 $J(0)$   $\gamma'(0)$   $J'(0)$   $\gamma'(0)$   $J'(0) = 0$   
 :  $f(s) = \langle J(s), \gamma'(s) \rangle$  1(a)  $f'' + K(s)f = 0$   $f(0) = 0$   $f'(0) =$   
 $\langle J'(0), \gamma'(0) \rangle = 0$  ODE  $f(s) \equiv 0$   $\langle J(s), \gamma'(s) \rangle = 0$   
 Bonnet O. Bonnet (1850, 1851) Hopf-Rinow  $K$   $K$   
 E. Calabi (1967), R. Schneider (1972)  
 $K \geq 1$   $\rho \leq \pi$  Bonnet  $\rho \leq \pi/\sqrt{\delta}$   $S$

## 5-4

5-4, 1 — do Carmo, Exercise 5-4, 1 Bonnet  $S$   $\rho \leq \pi/\sqrt{\delta}$   $K \geq \delta$   
 : Bonnet  $K \geq \delta > 0 \Rightarrow \rho \leq \pi/\sqrt{\delta}$   $\Rightarrow$  " " " "  
 —  
 : torus  $\rho \leq \pi$   $K$  0  
 $K \geq \delta > 0$  cap  $\leq \pi$   $K$  0 Bonnet

1.4. FIRST AND SECOND VARIATIONS OF ARC LENGTH; BONNET'S THEOREM 15

5-4, 2\* — Kazdan-Warner Remark, do Carmo, Exercise 5-4, 2  $S = \{z = f(x, y); (x, y) \in R^2\}$

$$\lim_{r \rightarrow \infty} \left( \inf_{x^2 + y^2 \geq r} K(x, y) \right) \leq 0.$$

: " "  $K$  Bonnet —  
 $S = \{z = f(x, y)\}$   $\liminf_{r \rightarrow \infty} K > 0$   $\delta > 0$   $K \geq$   
 $\delta$   $v = (0, 0, 1)$   $z$   $h(x, y, z) = z = f(x, y)$  5-3  $10 |\text{grad } h| \leq 1$   
 $\text{grad } h$   $\alpha(t)$   $R$  Bonnet  
 $p_r = (r \cos \theta, r \sin \theta, f(r \cos \theta, r \sin \theta))$   $K(p_r) \geq \delta$   $q = (0, 0, f(0, 0))$   
 $p_r$   $\gamma_r$   $q$   $p_r$  5-4  $l(\gamma_r) > \pi/\sqrt{\delta}$   $r$   
 $d(q, p_r) \leq \pi/\sqrt{\delta}$   $S$  5-3  $2 S$   $\liminf_{r \rightarrow \infty} K \leq 0$  11  
 5-4, 3 — do Carmo, Exercise 5-4, 3 a b  $S$   $\gamma(s)$   $s \in R$   
 $S$   $d(s) = d(\gamma(s), p)$   $p$   $s_0$   $d(s_0) \leq d(s)$   $s$   $p$   $\gamma(s_0)$   
 $\gamma$  fig:fig5-14 c  $S$   $K \leq 0$   $s_0$  : (a) —  
 (b)  $d(s) = d(\gamma(s), p)$   $\gamma$   $d(s)$  (c)  
 $S$   $K \leq 0$   
 : (a)  $h : [0, l] \times (-\epsilon, \epsilon) \rightarrow S$  5-4 2 " "

$$L'(0) = - \int_0^l \langle A(s), V(s) \rangle ds + \langle \frac{\partial h}{\partial s}(l, 0), \frac{\partial h}{\partial t}(l, 0) \rangle - \langle \frac{\partial h}{\partial s}(0, 0), \frac{\partial h}{\partial t}(0, 0) \rangle.$$

$h$   $V(0) = V(l) = 0$   
 (b)  $S$   $\gamma : R \rightarrow S$   $p \in S$   $d(s) = d(\gamma(s), p)$  5-3  $4 d$   $S$   $d$   
 $d(s) \rightarrow \infty$   $|s| \rightarrow \infty$   $[-R, R]$   $d$   $s_0$   $\Gamma$   $p$   $\gamma(s_0)$   
 Hopf-Rinow  $\Gamma$   $\gamma$   $\Gamma$   $V$   $\Gamma$   $\gamma$   $V(s_0) \neq$   
 0  $0 = L'(0) = - \int \langle A, V \rangle$   $A = 0$   $\Gamma$   $\Gamma$   $\gamma(s_0)$   $s_0$   
 $\gamma$   $d(s)$   $s_0$   
 (c)  $S$   $K \leq 0$   $\exp_p : T_p(S) \rightarrow S$  5-5  $S$   $R^2 \exp_p$   
 $\exp_p(v_1) = \exp_p(v_2)$   $p$   $v_1 = v_2$   $s_0$   $p$   $\gamma(s)$   
 5-4, 4\* — do Carmo, Exercise 5-4, 4  $y = y(x)$   $x \in [x_1, x_2]$   $y(x, t)$   
 $y(x, 0) = y(x)$   $y(x_1, t) = y(x_1)$   $y(x_2, t) = y(x_2)$   $I(t) = \int_{x_1}^{x_2} F(x, y(x, t), y'(x, t)) dx$  a  
 $y(x)$   $I(t)$   $dI/dt = 0$   $\int_{x_1}^{x_2} \eta(F_y - \frac{d}{dx} F_{y'}) dx = 0$   $\eta(x_1) = \eta(x_2) = 0$   
 $\eta$  b Euler-Lagrange  $F_y - \frac{d}{dx} F_{y'} = 0$  c  $F$   $x$   $F = F(y, y')$   
 $y' F_{y'} - F = \text{const}$  : " " " " Euler-Lagrange  
 $F$   $x$   
 : (a)  $y = y(x)$   $I(t)$   $dI/dt = 0$

$$\frac{dI}{dt} = \int_{x_1}^{x_2} \left( F_y \frac{\partial y}{\partial t} + F_{y'} \frac{\partial y'}{\partial t} \right) dx.$$

$$\eta(x) = (\partial y / \partial t)(x, 0) \quad \eta(x_1) = \eta(x_2) = 0$$

$$\int_{x_1}^{x_2} F_{y'} \eta' dx = [F_{y'} \eta]_{x_1}^{x_2} - \int_{x_1}^{x_2} \frac{d}{dx} F_{y'} \cdot \eta dx = - \int_{x_1}^{x_2} \frac{d}{dx} F_{y'} \cdot \eta dx,$$

$$\eta(x_1) = \eta(x_2) = 0$$

$$0 = \frac{dI}{dt} \Big|_{t=0} = \int_{x_1}^{x_2} \eta \left( F_y - \frac{d}{dx} F_{y'} \right) dx.$$

$$\begin{aligned} \eta \quad F_y - \frac{d}{dx} F_{y'} &= 0 \quad x \in [x_1, x_2] \\ \text{(b) Euler-Lagrange} \\ \text{(c) } F &= F(y, y') \end{aligned}$$

$$\frac{d}{dx} (y' F_{y'} - F) = y'' F_{y'} + y' ((F_y) y' + (F_{y'}) y'') - (F_y y' + F_{y'} y'') = y'' F_{y'} + y' F_y y' + F_{y'} y'' - F_y y' - F_{y'} y'' =$$

$$\begin{aligned} F_y &= (d/dx) F_{y'} \quad \text{Euler-Lagrange} \quad y' F_{y'} - F = \text{const} \\ 5-4, 5^* &\text{--- do Carmo, Exercice 5-4, 5 a} \quad S \quad y = f(x) \quad x \in [x_1, x_2] \quad x \quad S \quad y = \frac{1}{c} \cosh(cx + c_1) \quad \text{b} \quad \mathbf{x}(u, v) = \\ (f(v) \cos u, f(v) \sin u, g(v)) &\quad \text{Clairaut}^{12} \quad : \quad I = \int y \sqrt{1 + (y')^2} dx \\ \text{Euler-Lagrange} \quad y / \sqrt{1 + (y')^2} &= 1/c \quad y = \frac{1}{c} \cosh(cx + c_1) \\ \text{Clairaut} \end{aligned}$$

$$: \text{ (a) } \quad S \quad y = f(x) \quad x \in [x_1, x_2] \quad x$$

$$A(f) = 2\pi \int_{x_1}^{x_2} f(x) \sqrt{1 + (f'(x))^2} dx.$$

$$F(y, y') = y \sqrt{1 + (y')^2} \quad \text{Euler-Lagrange}$$

$$F_y - \frac{d}{dx} F_{y'} = 0 \quad \implies \quad \sqrt{1 + (y')^2} - \frac{d}{dx} \left( \frac{yy'}{\sqrt{1 + (y')^2}} \right) = 0.$$

$$\frac{y}{\sqrt{1 + (y')^2}} = c \quad (const),$$

$$y^2 = c^2(1 + (y')^2) \quad y' = \pm \sqrt{y^2/c^2 - 1}$$

$$y = \frac{1}{c} \cosh(cx + c_1), \quad c_1 = const.$$

$$\begin{aligned} \text{catenoid} \\ \text{(b) } \quad S \quad E &= f(v)^2 \quad F = 0 \quad G = (f')^2 + (g')^2 = 1 \quad v \end{aligned}$$

$$L = \int \sqrt{f(v)^2 (u')^2 + 1} dv, \quad u' = \frac{du}{dv}.$$

$$F = f^2(u')^2 \quad u \quad \text{(c)}$$

$$u' F_{u'} - F = \frac{f^2(u')^2}{\sqrt{f^2(u')^2 + 1}} - \sqrt{f^2(u')^2 + 1} = -\frac{1}{\sqrt{f^2(u')^2 + 1}} = c \quad (const),$$

$$f(v)^2 u' = \frac{c}{\sqrt{1 - c^2/f(v)^2}} = c \sqrt{\frac{f(v)^2}{f(v)^2 - c^2}}.$$

$$\text{Clairaut} \quad c \quad \text{Clairaut}$$

$$^{12} \text{Clairaut} \quad u' = c/(f^2) \sqrt{(f'^2 + g'^2)(f^2 - c^2)}^{-1} \cdot \sqrt{\dots}$$



## 1.5 Jacobi Fields and Conjugate Points

**Jacobi** .  $\gamma$  ——— ” ”  
 [Jacobi]  $\gamma : [0, l] \rightarrow S$   $h : [0, l] \times (-\epsilon, \epsilon) \rightarrow S$   $\gamma$   $t$   
 $h_t(s) = h(s, t)$   $J(s) = (\partial h / \partial t)(s, 0)$   $\gamma$  **Jacobi** Jacobi field  
**Trivial** .  $\gamma'(s)$  Jacobi  $h(s, t) = \gamma(s + t)$   
 $\gamma$  Jacobi  $J(s)$  Jacobi

$$\frac{D}{ds} \frac{D}{ds} J(s) + K(s)(\gamma'(s) \wedge J(s)) \wedge \gamma'(s) = 0,$$

$$K(s) S \gamma(s) \\ h_t(s) (D/\partial s)(\partial h / \partial s) = 0 \quad t \quad ?? \quad 5-4$$

$$0 = \frac{D}{\partial t} \frac{D}{\partial s} \frac{\partial h}{\partial s} = \frac{D}{\partial s} \frac{D}{\partial t} \frac{\partial h}{\partial s} + K \frac{\partial h}{\partial s} \wedge \frac{\partial h}{\partial t} \wedge \frac{\partial h}{\partial s}.$$

$$t = 0 \quad \text{Jacobi} \quad 13$$

$$\text{Jacobi} \quad 4$$

**Example** **Jacobi** .  $S^2$   $\theta = \pi/2$   $\gamma(s)$   $s = \varphi - \varphi_0$   $0 \leq s \leq \pi$   
 $\gamma$   $w(s)$

$$J(s) = (\sin s) w(s)$$

Jacobi  $K = 1$   $D^2 J / ds^2 + K J = -\sin s + \sin s = 0$   $J(0) = J(\pi) = 0$   
 [ ]  $\gamma : [0, l] \rightarrow S$   $\gamma(0) = p$   $q = \gamma(s_0)$   $p$   $\gamma$  conjugate Jacobi  
 $J(s)$   $\gamma$   $J(0) = J(s_0) = 0$

$$\cdot \quad \text{” ”} \quad p \quad p \quad \text{fig:fig5-18}$$

$$\cdot \quad v = l\gamma'(0) \in T_p(S) \quad q = \exp_p(v) \quad p \quad v \quad \exp_p : T_p(S) \rightarrow S$$

5

$$p, q \in S \quad \gamma : [0, l] \rightarrow S \quad p = \gamma(0) \quad q = \exp_p(l\gamma'(0)) \quad q \quad p \quad \gamma \\ v = l\gamma'(0) \quad \exp_p$$

$$S \quad K \leq 0 \quad p \in S \quad p \quad \text{—} \quad K \leq 0$$

$$\text{Jacobi} \quad J(s) \quad \gamma \quad J(0) = J(l) = 0 \quad 5-5 \quad \langle J(s), \gamma'(s) \rangle = 0 \quad \text{Jacobi}$$

$$K \leq 0$$

$$\frac{d}{ds} \left\langle \frac{DJ}{ds}, J \right\rangle = \left| \frac{DJ}{ds} \right|^2 - K|J|^2 \geq 0.$$

$$\langle DJ/ds, J \rangle \quad s = 0 \quad s = l \quad J(0) = J(l) = 0 \quad \langle DJ/ds, J \rangle \equiv 0 \\ |J|^2 = \text{const} \quad J(0) = 0 \quad J \equiv 0 \quad 14$$

$$\cdot \quad K \leq 0 \quad \exp_p : T_p(S) \rightarrow S$$

$$\textbf{Example.} \quad K = 0 \quad K < 0 \quad K = 0$$

$$[\text{Gauss}] \quad p \in S \quad \mathbf{u} \in T_p(S) \quad \mathbf{w} \in (T_p(S))_{\mathbf{u}}$$

$$\langle \mathbf{u}, \mathbf{w} \rangle = \langle (d\exp_p)_{\mathbf{u}}(\mathbf{u}), (d\exp_p)_{\mathbf{u}}(\mathbf{w}) \rangle.$$

· ” ”

Jacobi

conjugate locus fig:fig5-19

L. Green

5-5 Jacobi 5-6 5-8

<sup>13</sup> sec:derivation-jacobi-equation

<sup>14</sup> sec:derivation-no-conjugate

[Figure 5-11][width=]../../PDFs/differential-geometry/transcript/Do Carmo - Differential Geometry of Curves and Surfaces/hybrid<sub>ocr</sub>/images/37468c0aac1e027f8629b5ba8f338edfaa0a97489c

[Figure 5-13.][width=]../../PDFs/differential-geometry/transcript/Do Carmo - Differential Geometry of Curves and Surfaces/hybrid<sub>ocr</sub>/images/3b2c5faa763a8cefdddf8608839bb7df8d3e556c44e5813

[Figure 5-14. (c)][width=]../../PDFs/differential-geometry/transcript/Do Carmo - Differential Geometry of Curves and Surfaces/hybrid<sub>ocr</sub>/images/b0507eebec0f2e639ed7f70f73f1aeae139ba72e68df672

[Figure 5-18. Jacobi  $J(s) = \sin s$   $w(s)$   $J(\pi) = 0$ ][width=]../../PDFs/differential-geometry/transcript/Do Carmo - Differential Geometry of Curves and Surfaces/hybrid<sub>ocr</sub>/images/8d4591e058e498f71e52247b45c12013977e762221b6794

[Figure 5-19.][width=]../../PDFs/differential-geometry/transcript/Do Carmo - Differential Geometry of Curves and Surfaces/hybrid<sub>ocr</sub>/images/00ba2e82b8c9203c068529df1f226052e698011e2f85a9c

[Figure 5-16.][width=]../../PDFs/differential-geometry/transcript/Do Carmo - Differential Geometry of Curves and Surfaces/hybrid<sub>ocr</sub>/images/0c5380d22284c20ad43fd260afcb6dbe28a06d22eb7ccc

[Figure 5-17. Jacobi  $J(s) = s(d \exp_p)_{sv}(w)$ ][width=]../../PDFs/differential-geometry/transcript/Do Carmo - Differential Geometry of Curves and Surfaces/hybrid<sub>ocr</sub>/images/ccf3838c806db8e32228ed9891f30b1014df2cdb711c624e

[Figure 5-20. Gauss][width=]../../PDFs/differential-geometry/transcript/Do Carmo - Differential Geometry of Curves and Surfaces/hybrid<sub>ocr</sub>/images/1ba2e22f1f13d88ddd6d0d38ec454df7a18ff5cc9e35a9

## 5-6

## 1.6 Covering Spaces; The Theorems of Hadamard

$$K \leq 0 \quad \exp_p : T_p(S) \rightarrow S$$

## A.

[ ]  $\tilde{B}, B \subset R^3 \quad \pi : \tilde{B} \rightarrow B \quad \text{covering map}$

$$1. \quad \pi^{-1}(\pi(\tilde{B})) = \tilde{B}$$

$$2. \quad p \in B \quad U \text{ distinguished neighborhood} \quad \pi^{-1}(U) = \bigcup_{\alpha} V_{\alpha} \quad V_{\alpha} \\ \pi|_{V_{\alpha}} : V_{\alpha} \rightarrow U$$

$\tilde{B} \rightarrow B$  covering space

**Example 1** .  $P \quad \pi(u, v) = (\cos u, \sin u, v) \quad S = \{(x, y, z) : x^2 + y^2 = 1\}$  fig:fig5-22 " "

**Example 2** .  $H = \{(\cos t, \sin t, bt) : t \in R\} \quad \pi(x, y, z) = (x, y, 0)$   $S^1$  fig:fig5-23

**Example 4** .  $k \quad \pi(\cos t, \sin t) = (\cos kt, \sin kt) \quad k$  " "

$$\pi : \tilde{B} \rightarrow B \quad \alpha : [0, l] \rightarrow B \quad \tilde{p}_0 \in \tilde{B} \quad \pi(\tilde{p}_0) = \alpha(0) \quad \tilde{\alpha} : [0, l] \rightarrow \tilde{B} \\ \tilde{\alpha}(0) = \tilde{p}_0 \quad \pi \circ \tilde{\alpha} = \alpha$$

[ ]  $B \subset R^3 \quad \text{simply connected}$   
= " "

$$\pi : \tilde{B} \rightarrow B \quad \tilde{p}_1, \tilde{p}_2 \in \tilde{B} \quad \pi(\tilde{p}_1) = \pi(\tilde{p}_2) = p \in B \quad \tilde{p}_1 \sim \tilde{p}_2 \quad \tilde{\alpha}_0 \quad \alpha_0 = \pi \circ \tilde{\alpha}_0 \\ \alpha_0 : [0, l] \rightarrow B \quad \tilde{\alpha}_0 : [0, l] \rightarrow \tilde{B} \quad \tilde{p}_1 = \tilde{p}_2^{15}$$

## B. Hadamard

$$S \quad \exp_p : T_p(S) \rightarrow S \quad K \leq 0 \quad \exp_p : T_p(S) \rightarrow S \quad \mathbf{u}, \mathbf{w} \in T_p(S)$$

$$\langle (d \exp_p)_u(\mathbf{w}), (d \exp_p)_u(\mathbf{w}) \rangle \geq \langle \mathbf{w}, \mathbf{w} \rangle.$$

$$[16] \quad K \leq 0 \quad \text{Jacobi} \quad ?? \text{ Gauss } \quad \text{Jacobi} \quad |J(s)|^2 \geq s^2 |w|^2 \quad s = |u|$$

$$K \equiv 0 \quad \exp_p : T_p(S) \rightarrow S \\ S \quad K \leq 0 \quad \exp_p : T_p(S) \rightarrow S$$

$$\exp_p \quad \text{Hadamard} \quad S \quad K \leq 0 \quad \exp_p : T_p(S) \rightarrow S \quad S$$

<sup>15</sup> sec:derivation-covering-homeomorphism

<sup>16</sup> sec:derivation-length-increasing

<sup>17</sup> sec:derivation-exp-covering

$\exp_p : T_p(S) \rightarrow S$   
 $[ \text{Hadamard} ] \quad S \quad \text{Gauss} \quad N : S \rightarrow S^2 \quad S$   
 $K > 0 \quad N \quad dN_p \quad \Rightarrow \quad N \quad S^2 \quad N \quad + \quad \Rightarrow \quad 18$   
 $\text{Hadamard} \quad \text{Gauss} \quad S \quad \text{---}$   
  
 $\text{Hadamard} \quad \text{Klingenberg} \quad 8 \quad 9$   
 $\text{Hadamard} \quad K \leq 0 \quad K > 0$   
 $\text{Hadamard} \quad \text{complete surfaces}$   
 $\text{fig:fig5-30} \quad \text{fig:fig5-43}$

## 1.7 Global Theorems for Curves: The Fary-Milnor Theorem

$\text{degree}$   
 $S^1 \quad \cdot \quad \pi : R \rightarrow S^1 \quad \pi(x) = (\cos x, \sin x) \quad \varphi : S^1 \rightarrow S^1 \quad \varphi : [0, l] \rightarrow$   
 $\tilde{\varphi} : [0, l] \rightarrow R \quad \tilde{\varphi}(l) - \tilde{\varphi}(0) = 2\pi \cdot \deg \varphi \quad \deg \varphi \quad \varphi \quad \text{---} \quad \varphi \quad S^1 \quad \text{" "}$   
 $\cdot$   
 $\cdot \quad \alpha : [0, l] \rightarrow R^2 \quad \varphi(t) = \alpha'(t)/|\alpha'(t)| \quad \text{rotation index}$   
 $\text{dex}$   
 $[ \quad \text{Jordan} \quad ] \quad \alpha : [0, l] \rightarrow R^2 \quad R^2 - \alpha([0, l]) \quad \alpha([0, l])$   
 $[ \quad ] \quad \beta : [0, l] \rightarrow R^2 \quad \beta \quad \pm 1$   
 $\cdot$   
 $k$   
 $k \Leftrightarrow \pm 1 \Leftrightarrow$   
**Fenchel**  
 $[ \text{Fenchel} ] \quad \geq 2\pi$   
 $\cdot \quad T \quad \alpha \quad r \quad r \quad R \subset T \quad K \geq 0$

$$\int_0^l K \, ds = 2 \int_0^l k(s) \, ds = 2 \cdot (\text{rotation index}).$$

$R \quad \int_0^l K \, ds \geq 4\pi \quad \geq 2\pi \quad 19$   
**Fary-Milnor**  
 $[ \text{Fary-Milnor} ] \quad \text{knotted} \quad > 4\pi$   
 $2\pi \quad > 4\pi \quad \text{" "}$   
 $\text{Fenchel} \quad \text{Fary-Milnor} \quad \text{" "}$   
 $\text{Fary-Milnor} \quad > 4\pi$   
 $\text{Fenchel} \quad \text{Fenchel} \quad 1929 \quad \text{Fary} \quad \text{Milnor} \quad > 4\pi$   
 $?? \quad \text{Gauss-Bonnet} \quad \pm 1$   
 $?? \quad \text{Jordan} \quad \text{Jordan}$   
 $?? \quad \text{Gauss-Bonnet}$   
 $\text{ellipsoid} \quad \text{fig:fig5-19}$   
 $5-7 \quad 5-6 \quad A \quad \text{Hadamard}$

<sup>18</sup> sec:derivation-hadamard-second  
<sup>19</sup> sec:derivation-fenchel

## 1.7. GLOBAL THEOREMS FOR CURVES: THE FARY-MILNOR THEOREM 21

[Figure 5-22.

][width=]../../PDFs/differential-  
 geometry/transcript/Do Carmo -  
 Differential Geometry of Curves and  
 Surfaces/hybrid<sub>o</sub>cr/images/e3bfd94692ad63e4461772c4d8ced89c2c53cbca64ab6e61c47a13e7b6

[Figure 5-23.

][width=]../../PDFs/differential-  
 geometry/transcript/Do Carmo -  
 Differential Geometry of Curves and  
 Surfaces/hybrid<sub>o</sub>cr/images/ac958466d6b11de3d23faca24b2240c942baf95a7a3335bc0919820123

[Figure 5-24.  $1 \quad \pi \quad U$

][width=]../../PDFs/differential-  
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 Differential Geometry of Curves and  
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[Figure 5-25.

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 Surfaces/hybrid<sub>o</sub>cr/images/df62e07203a43bf2814074f03ecdbc2613cda45f278670c87e811bc44

[Figure 5-26.

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 of Curves and Surfaces/hybrid<sub>o</sub>cr/images/550e844cd15958bac510049e52008452c4a2c648f23bb490087e89b11af0b17859fc4e77

[Figure 5-

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[Figure 5-27.

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 Surfaces/hybrid<sub>o</sub>cr/images/2484e82cc1d0aca25cf6b5624d150b5db441be552864e152f42880a62d

[Figure 5-28.  $\tilde{H} \quad H$

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[Figure 5-29. Klingenberg

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[Figure 5-12. Osserman  $\theta^{-1} \eta_0$   
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[Figure 5-21.  
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[Figure 5-10. Hopf-Rinow  
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[Figure 5-4.  $p \in \text{Bd } S$   
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[Figure 5-8.  $S^2 - \{p\}$   
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[Figure 5-6.  
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[Figure 5-30.  
 $\varphi : S^1 \rightarrow S^1$   
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[Figure 5-31.  
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[Figure 5-  
 31b][width=]../../PDFs/differential-  
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[Figure 5-32. Jordan  
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[Figure 5-34.  
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[Figure 5-36.  
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[Figure 5-37. Fenchel  
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[Figure 5-38.  
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[Figure 5-38b][width=]../../PDFs/differential-  
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[Figure 5-39. Fary-Milnor  
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[Figure 5-41. Schur  
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[Figure 5-42. Stoker  
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[Figure 5-40.  
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[Figure 5-40b][width=]../../PDFs/differential-  
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[Figure 5-40c][width=]../../PDFs/differential-  
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[Figure 5-43.  
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|             |                        |                                        |                                                                                                          |                           |                                        |
|-------------|------------------------|----------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------|
|             |                        | Gauss-Bonnet                           |                                                                                                          | 4-5                       |                                        |
|             | ??                     |                                        | " "                                                                                                      |                           |                                        |
|             |                        | 5-6                                    | A                                                                                                        |                           |                                        |
| Fenchel     |                        |                                        |                                                                                                          | $2\pi$                    |                                        |
|             | ??                     | " > 4π"                                |                                                                                                          | $2 \int k ds$             | $\geq 2$                               |
|             |                        | Fary-Milnor                            |                                                                                                          | _____                     |                                        |
| 5-7         |                        | Stoker                                 |                                                                                                          |                           |                                        |
|             | Stoker                 |                                        |                                                                                                          | $\int_0^l \tau(s) ds = 0$ |                                        |
| Schur       | 7                      |                                        |                                                                                                          | $k \geq \tilde{k}$        | $d \geq \tilde{d}$ " "                 |
|             | _____                  | _____                                  |                                                                                                          |                           |                                        |
| fig:fig5-30 | fig:fig5-42            |                                        |                                                                                                          | Fenchel                   | Fary-Milnor                            |
| 5-7         | 5-6                    | A                                      |                                                                                                          | $R \rightarrow S^1$       | Jordan Gauss-Bonnet                    |
|             |                        |                                        |                                                                                                          |                           |                                        |
| 5-7         |                        |                                        |                                                                                                          |                           |                                        |
|             | Hadamard               | ?? ??                                  |                                                                                                          | $K \leq 0$                | $K > 0$                                |
|             |                        |                                        |                                                                                                          |                           |                                        |
| 5-7         | 5-5                    | Jacobi                                 | 5-4                                                                                                      | Fenchel                   | Jacobi                                 |
| Bonnet      | ??                     |                                        |                                                                                                          | ??                        | $V(s) = w(s) \sin \frac{\pi}{l} s$ " " |
|             |                        |                                        |                                                                                                          |                           |                                        |
|             | Hopf-Rinow             | ??                                     |                                                                                                          |                           |                                        |
|             | ??                     |                                        | ?? Mainardi-Codazzi                                                                                      |                           |                                        |
|             |                        | +                                      | / / $\Rightarrow$                                                                                        |                           |                                        |
|             | 5-3                    | $\rightarrow$ 5-4                      | Bonnet $\rightarrow$ 5-5                                                                                 | Jacobi $\rightarrow$ 5-6  | Hadamard                               |
| 5-8         | 5-11                   | Hilbert                                |                                                                                                          |                           |                                        |
|             | 5-2                    | 5-7                                    |                                                                                                          |                           |                                        |
| 5-10        |                        | $R^3$                                  |                                                                                                          |                           |                                        |
| Hilbert     | ??                     | " "                                    | $R^3$                                                                                                    | $R^3$                     | _____                                  |
|             |                        |                                        |                                                                                                          |                           |                                        |
|             | *                      | 5-3                                    | 11 Osserman                                                                                              | _____                     | 5-5 3 _____                            |
| Jacobi      |                        |                                        |                                                                                                          |                           |                                        |
|             | 5-7, 1                 | — do Carmo, Exercise 5-7, 1            | Determine the rotation indices of curves                                                                 |                           |                                        |
|             | (a), (b), (c), and (d) | in Fig. 5-40.                          | :                                                                                                        | rotation index            |                                        |
|             | $\pm 1$                |                                        |                                                                                                          |                           |                                        |
|             | :                      | fig:fig5-40                            |                                                                                                          |                           |                                        |
|             | • (a)                  | +1                                     | -1                                                                                                       |                           |                                        |
|             | • (b)                  | 0                                      | 2                                                                                                        |                           |                                        |
|             | • (c)                  | 2                                      |                                                                                                          |                           |                                        |
|             | • (d)                  | 1                                      |                                                                                                          |                           |                                        |
|             |                        | $\varphi(t) = \alpha'(t)/ \alpha'(t) $ | $\pm 1$                                                                                                  |                           |                                        |
|             | 5-7, 2                 | — do Carmo, Exercise 5-7, 2            | Let $\alpha(t) = (x(t), y(t))$ , $t \in [0, l]$ , be a                                                   |                           |                                        |
|             |                        |                                        | differentiable plane closed curve. Let $p_0 = (x_0, y_0) \in R^2$ , $(x_0, y_0) \notin \alpha([0, l])$ , |                           |                                        |
|             |                        |                                        | and define the functions                                                                                 |                           |                                        |

$$a(t) = \frac{x(t) - x_0}{\{(x(t) - x_0)^2 + (y(t) - y_0)^2\}^{1/2}},$$

$$b(t) = \frac{y(t) - y_0}{\{(x(t) - x_0)^2 + (y(t) - y_0)^2\}^{1/2}}.$$

a. Use Lemma 1 of Sec. 4-4 to show that the differentiable function

$$\varphi(t) = \varphi_0 + \int_0^t (ab' - ba') dt, \quad a' = \frac{da}{dt}, b' = \frac{db}{dt},$$

is a determination of the angle that the  $x$  axis makes with the position vector  $(\alpha(t) - p_0)/|\alpha(t) - p_0|$ .

b. Use part a to show that when  $\alpha$  is a differentiable closed plane curve, the winding number of  $\alpha$  relative to  $p_0$  is given by the integral

$$w = \frac{1}{2\pi} \int_0^l (ab' - ba') dt.$$

$$\begin{array}{ll} \text{: winding number} & w = \frac{1}{2\pi} \int (a db' - b da') \\ \text{: (a) 1 Sec. 4-4} & \varphi'(t) = a b' - b a' \quad \varphi(t) = (\alpha(t) - p_0) \cdot x \\ \text{(b)} & w = \frac{\varphi(l) - \varphi(0)}{2\pi} \end{array}$$

$$w = \frac{\varphi(l) - \varphi(0)}{2\pi} = \frac{1}{2\pi} \int_0^l (a b' - b a') dt.$$

5-7, 3 — do Carmo, Exercise 5-7, 3 Let  $\alpha: [0, l] \rightarrow R^2$  and  $\beta: [0, l] \rightarrow R^2$  be two differentiable plane closed curves, and let  $p_0 \in R^2$  be a point such that  $p_0 \notin \alpha([0, l])$  and  $p_0 \notin \beta([0, l])$ . Assume that, for each  $t \in [0, l]$ , the points  $\alpha(t)$  and  $\beta(t)$  are closer than the points  $\alpha(t)$  and  $p_0$ ; i.e.,

$$|\alpha(t) - \beta(t)| < |\alpha(t) - p_0|.$$

Use Exercise 2 to prove that the winding number of  $\alpha$  relative to  $p_0$  is equal to the winding number of  $\beta$  relative to  $p_0$ .

$$\begin{array}{ll} \text{:} & w = \frac{1}{2\pi} \int (a b' - b a') dt \\ \text{: (a)} & u_\alpha(t) = \frac{\alpha(t) - p_0}{|\alpha(t) - p_0|} \quad u_\beta(t) = \frac{\beta(t) - p_0}{|\beta(t) - p_0|} \quad w(\alpha) = w(\beta) \end{array}$$

5-7, 4 — do Carmo, Exercise 5-7, 4 a. Let  $C$  be a regular plane closed convex curve. Since  $C$  is simple, it determines, by the Jordan curve theorem, an interior region  $K \subset R^2$ . Prove that  $K$  is a convex set (i.e., given  $p, q \in K$ , the segment of straight line  $\overline{pq}$  is contained in  $K$ ; cf. Exercise 9, Sec. 1-7).

b. Conversely, let  $C$  be a regular plane curve (not necessarily closed), and assume that  $C$  is the boundary of a convex region. Prove that  $C$  is convex.

$$\begin{array}{ll} \text{:} & \text{(a) } C \quad K \quad p, q \in K \quad L \quad \overline{pq} \quad LK \quad L \quad C \quad \text{Jordan} \quad p \\ & q \quad C \quad L \subseteq K \quad K \\ \text{(b)} & C \quad D \quad C \quad a, b \in C \quad \overline{ab} \cap D \quad c \in \overline{ab} \quad C \quad C \end{array}$$

5-7, 5 — do Carmo, Exercise 5-7, 5 Let  $C$  be a regular plane, closed, convex curve. By Exercise 4, the interior of  $C$  is a convex set  $K$ . Let  $p_0 \in K$ ,  $p_0 \notin C$ .

a. Show that the line which joins  $p_0$  to an arbitrary point  $q \in C$  is not tangent to  $C$  at  $q$ . b. Conclude from part a that the rotation index of  $C$  is equal to the winding number of  $C$  relative to  $p_0$ . c. Obtain from part b a simple proof for the fact that the rotation index of a closed convex curve is  $\pm 1$ .

$p_0$   $p_0$  4  
: (a) 4  $p_0$   $C$   $q$   $C$   $q$   $C$   $p_0$   
(b) (a)  $C$   $p_0$   
(c) (b) 2  $\pm 1$

5-7, 6 — do Carmo, Exercise 5-7, 6 Let  $\alpha: [0, l] \rightarrow R^3$  be a regular closed curve parametrized by arc length. Assume that  $0 \neq |k(s)| \leq 1$  for all  $s \in [0, l]$ . Prove that  $l \geq 2\pi$  and that  $l = 2\pi$  if and only if  $\alpha$  is a plane convex curve.

Fenchel  $\geq 2\pi$   $|k| \leq 1$   $l$   $l \geq 2\pi$   
:  $\int_0^l |k(s)| ds$   $|k(s)| \leq 1$   $\int_0^l |k(s)| ds \leq l$  Fenchel ??  $\geq 2\pi$   
 $l \geq 2\pi$   $l = 2\pi$   $|k(s)| \equiv 1$  Fenchel  $|k| \leq 1$   $k > 0$   
 $l = 2\pi$

5-7, 7 — do Carmo, Exercise 5-7, 7 (Schur's Theorem for Plane Curves) Let  $\alpha: [0, l] \rightarrow R^2$  and  $\tilde{\alpha}: [0, l] \rightarrow R^2$  be two plane convex curves parametrized by arc length, both with the same length  $l$ . Denote by  $k$  and  $\tilde{k}$  the curvatures of  $\alpha$  and  $\tilde{\alpha}$ , respectively, and by  $d$  and  $\tilde{d}$  the lengths of the chords of  $\alpha$  and  $\tilde{\alpha}$ , respectively; i.e.,

$$d(s) = |\alpha(s) - \alpha(0)|, \quad \tilde{d}(s) = |\tilde{\alpha}(s) - \tilde{\alpha}(0)|.$$

Assume that  $k(s) \geq \tilde{k}(s)$ ,  $s \in [0, l]$ . We want to prove that  $d(s) \leq \tilde{d}(s)$ ,  $s \in [0, l]$  (i.e., if we stretch a curve, its chords become longer) and that equality holds for  $s \in [0, l]$  if and only if the two curves differ by a rigid motion. We remark that the theorem can be extended to the case where  $\tilde{\alpha}$  is a space curve and has a number of applications, Compare S. S. Chern [10].

The following outline may be helpful.

a. Fix a point  $s = s_1$ . Put both curves  $\alpha(s) = (x(s), y(s))$ ,  $\tilde{\alpha}(s) = (\tilde{x}(s), \tilde{y}(s))$  in the lower half-plane  $y \leq 0$  so that  $\alpha(0), \alpha(s_1), \tilde{\alpha}(0)$ , and  $\tilde{\alpha}(s_1)$  lie on the  $x$  axis and  $x(s_1) > x(0)$ ,  $\tilde{x}(s_1) > \tilde{x}(0)$  (see Fig. 5-41). Let  $s_0 \in [0, s_1]$  be such that  $\alpha'(s_0)$  is parallel to the  $x$  axis. Choose the function  $\theta(s)$  which gives a differentiable determination of the angle that the  $x$  axis makes with  $\alpha'(s)$  in such a way that  $\theta(s_0) = 0$ . Show that, by convexity,  $-\pi \leq \theta \leq \pi$ .

b. Let  $\tilde{\theta}(s)$ ,  $\tilde{\theta}(s_0) = 0$ , be a differentiable determination of the angle that the  $x$  axis makes with  $\tilde{\alpha}'(s)$ . (Notice that  $\tilde{\alpha}'(s_0)$  may no longer be parallel to the  $x$  axis.) Prove that  $\tilde{\theta}(s) \leq \theta(s)$  and use part a to conclude that

$$d(s_1) = \int_0^{s_1} \cos \theta(s) ds \leq \int_0^{s_1} \cos \tilde{\theta}(s) ds \leq \tilde{d}(s_1).$$

For the equality case, just trace back your steps and apply the uniqueness theorem for plane curves. : Schur  $k \geq \tilde{k}$   $d \leq \tilde{d}$   
" "

$$\begin{array}{l} : \quad (a) \quad s_0 \quad \alpha'(s_0) \quad x \quad -\pi \leq \theta \leq \pi \quad k \geq \tilde{k} \quad \theta' \geq \tilde{\theta}' \quad \theta(s_0) = \\ \tilde{\theta}(s_0) = 0 \quad \theta(s) \geq \tilde{\theta}(s) \end{array}$$

$$d(s_1) = \int_0^{s_1} \cos \theta \, ds \leq \int_0^{s_1} \cos \tilde{\theta} \, ds \leq \tilde{d}(s_1).$$

$$k \equiv \tilde{k}$$

5-7, 8 — do Carmo, Exercise 5-7, 8 (Stoker's Theorem for Plane Curves) Let  $\alpha: R \rightarrow R^2$  be a regular plane curve parametrized by arc length. Assume that  $\alpha$  satisfies the following conditions:

1. The curvature of  $\alpha$  is strictly positive. 2.  $\lim_{s \rightarrow \pm\infty} |\alpha(s)| = \infty$ ; that is, the curve extends to infinity in both directions. 3.  $\alpha$  has no self-intersections.

The goal of the exercise is to prove that the total curvature of  $\alpha$  is  $\leq \pi$ .

The following indications may be helpful. Assume that the total curvature is  $> \pi$  and that  $\alpha$  has no self-intersections. To obtain a contradiction, proceed as follows:

a. Prove that there exist points, say,  $p = \alpha(0)$ ,  $q = \alpha(s_1)$ ,  $s_1 > 0$ , such that the tangent lines  $T_p, T_q$  at the points  $p$  and  $q$ , respectively, are parallel and there exists no tangent line parallel to  $T_p$  in the arc  $\alpha([0, s_1])$ . b. Show that as  $s$  increases,  $\alpha(s)$  meets  $T_p$  at a point, say,  $r$  (Fig. 5-42). c. The arc  $\alpha((-\infty, 0))$  must meet  $T_p$  at a point  $t$  between  $p$  and  $r$ . d. Complete the arc  $tpqr$  of  $\alpha$  with an arc  $\beta$  without self-intersections joining  $r$  to  $t$ , thus obtaining a closed curve  $C$ . Show that the rotation index of  $C$  is  $\geq 2$ . Show that this implies that  $\alpha$  has self-intersections, a contradiction.

: Stoker  $\leq \pi$   
 $> \pi$

$$\begin{array}{l} : \quad > \pi \quad s_1 > 0 \quad \int_0^{s_1} k \, ds > \pi \quad (a) \quad p = \alpha(0) \quad q = \alpha(s_1) \quad T_p \quad T_q \\ T_p < \pi \quad (b) \quad \alpha \quad T_p \quad r \quad (c) \quad \alpha \quad (-\infty, 0) \quad T_p \quad t \quad p \quad r \\ \beta \quad t \quad r \quad C \quad C \quad \geq 2 \quad C \quad \geq 2 \quad \leq \pi \end{array}$$

5-7, 9\* — do Carmo, Exercise 5-7, 9 Let  $\alpha: [0, l] \rightarrow S^2$  be a regular closed curve on a sphere  $S^2 = \{(x, y, z) \in R^3; x^2 + y^2 + z^2 = 1\}$ . Assume that  $\alpha$  is parametrized by arc length and that the curvature  $k(s)$  is nowhere zero. Prove that

$$\int_0^l \tau(s) \, ds = 0.$$

$$\begin{array}{l} : \quad \alpha: [0, l] \rightarrow S^2 \quad k^2 + \tau^2 = 1 \quad dN/ds = -k\mathbf{t} + \tau\mathbf{b} \text{ Darboux} \\ N: S^2 \rightarrow S^2 \text{ Gauss} \quad N \circ \alpha: S^2 \quad N \quad S^2 \quad \int_0^l \tau \, ds = 0 \end{array}$$

## 1.8 Surfaces of Zero Gaussian Curvature

$$\begin{array}{l} 4-6 \quad R^3 \\ S \subset R^3 \quad K \equiv 0 \quad S \\ \cdot \quad p \in S \quad \text{generator} \end{array}$$

$P$  ———  $P, S$  planar points  $H = 0$   $U = S - P$  parabolic points  
 $P$  ———  $p \in U$   $p$  1 Mainardi-Codazzi  $k_1 =$   
 $0, k_2$   
 $p \in U$   $p$   
 $\begin{bmatrix} \cdot \\ \cdot \end{bmatrix}$   $v = \text{const}$   $e = 0$   $N_u = 0$  Mainardi-Codazzi  $\langle \mathbf{x}, N \rangle =$   
 $\varphi(v)$   $N_v \neq 0$  20  
**Example**  $\cdot$  fig:fig5-43  $K \equiv 0$   $ABC$   $P$   
 $S$   $R^3$   
Mainardi-Codazzi  $K = 0$  ———

5-6, 1 — do Carmo, Exercise 5-6, 1  $\pi : R \rightarrow S^1 = \{(x, y) \in R^2 : x^2 + y^2 = 1\}$   
 $\pi(t) = (\cos t, \sin t)$   $\pi : R \rightarrow S^1$   $2\pi$   $t$   $p \in S^1$   $U$   
 $\pi$  " "  $\pi$   $U$   
:

1.  $\pi(R) = S^1$

2.  $p = (\cos t_0, \sin t_0) \in S^1$   $U = \{(\cos t, \sin t) : |t - t_0| < \pi/2\}$   $\pi^{-1}(U) =$   
 $\bigcup_{k \in \mathbb{Z}} (t_0 + 2k\pi - \pi/2, t_0 + 2k\pi + \pi/2)$   $\pi$

5-6, 2 — do Carmo, Exercise 5-6, 2  $\pi : R^2 - \{(0, 0)\} \rightarrow R^2 - \{(0, 0)\}$   $\pi(x, y) =$   
 $(x^2 - y^2, 2xy)$   $\pi : \pi$   $z \mapsto z^2$   $(x, y)$   $x + iy$   $w$   $z$   $-z$   
 $\pi : z = x + iy$   $\pi(z) = z^2$   $w \neq 0$   $z^2 = w$   $z = \pm\sqrt{w}$   $\pi^{-1}(w) =$   
 $\{\sqrt{w}, -\sqrt{w}\}$   $w$   $U$   $U$   $\pi$

5-6, 3 — do Carmo, Exercise 5-6, 3  $S = \{(\cos t, \sin t, bt) : t \in R\}$   $L$   $z$   
 $\pi : S - L \rightarrow R^2 - \{(0, 0)\}$   $\pi(x, y, z) = (x, y)$   $\pi : S \rightarrow R^2 - \{(0, 0)\}$   
 $(x, y) \neq (0, 0)$   $\pi$   $(x, y)$   
 $\pi : (x, y, z) \in S - L$   $x^2 + y^2 = 1$   $z$   $S$   $(x, y) \neq$   
 $(0, 0)$   $\pi^{-1}(x, y) = S - L$   $(x, y)$   $U$   $U$   $U$   $z$   $\pi$   
 $\pi$

5-6, 4 — do Carmo, Exercise 5-6, 4  $\pi : C - \{0\} \rightarrow C - \{0\}$   $\pi(z) = z^n$   
 $n$  :  $n$   $w$   $n$   $z$   $n$   
 $\pi : w \neq 0$   $z^n = w$   $n$   $z_k = w^{1/n} e^{2\pi i k/n}$   $k = 0, \dots, n-1$   $\pi$   
 $n$   $\pi$   $n$

5-6, 5 — do Carmo, Exercise 5-6, 5  $B \subset R^3$

1.  $p, q \in B$   $\alpha_0 : [0, l] \rightarrow B$   $\alpha_1 : [0, l] \rightarrow B$   $B$   $\alpha_0$   $\alpha_1$

2.  $p \in B$   $\alpha : [0, l] \rightarrow B$   $\alpha(0) = \alpha(l) = p$   $\alpha$   $\alpha(s) = p$   $s$   
 $\alpha(s) = p$

: ———

: (1)  $\Rightarrow$  (2)  
(2)  $\Rightarrow$  (1):  $p, q$   $\alpha_0, \alpha_1$   $\beta$   $p$   $q$   $\alpha_0$   $\beta^{-1} * \alpha_1 * \beta$  (2)  $\alpha_0$   
 $\beta^{-1} * \alpha_0 * \beta$   $\alpha_0$   $\alpha_1$  (2)  $\alpha_0 \# \alpha_1^{-1}$

5-6, 6 — do Carmo, Exercise 5-6, 6  $p_0 \in R^2$   $\varphi_t(p) = tp_0 + (1-t)p$   $R^2$   
 $\varphi_t$   $p_0$  — homotopy  
 $\alpha : [0, 1] \rightarrow R^2$   $H(s, t) = \varphi_t(\alpha(s)) = tp_0 + (1-t)\alpha(s)$   $H(s, 0) = \alpha(s)$   $H(s, 1) = p_0$   $H$   $\alpha$   $R^2$   
 5-6, 7a-b — do Carmo, Exercise 5-6, 7 a  $S^2$   $S^2$   $S^2$   $S^2$   
 c (a)(b)  $S^2$   $S^2$   $R^2$   $R^2$   
 $S^2$   $R^2$   
 $\psi : S^2 - \{N\} \rightarrow R^2$   $\alpha : S^2 - \{N\} \rightarrow R^2$   $\psi \circ \alpha : R^2$   
 6  $\alpha : S^2 - \{N\} \rightarrow R^2$   
 (b)  $\alpha : S^2 \rightarrow R^2$   $\alpha : S^2 \rightarrow R^2$   $\alpha : D \rightarrow R^2$   $\alpha : R^2 \rightarrow R^2$   
 $\alpha : p \rightarrow R^2$   
 (c) (a)(b)  $S^2$  (a)  $S^2$   
 5-6, 8\* Klingenberg — do Carmo, Exercise 5-6, 8  $S \subset R^3$   $K \leq 0$   
 $K_0 \geq 0$   $p, q \in S$   $\gamma_0$   $\gamma_1$   $p$   $q$   $l(\gamma_0) \leq l(\gamma_1)$   $\gamma_0$   $\gamma_1$   
 $t_0 \in [0, 1]$

$$l(\gamma_0) + l(\alpha_{t_0}) \geq \frac{2\pi}{\sqrt{K_0}}.$$

” ” fig:fig5-29  $l(\gamma_0) < \pi/\sqrt{K_0}$

(a) 5-5 3  $\exp_p : T_p(S) \rightarrow S$   $p$   $\pi/\sqrt{K_0}$   $B$

(b)  $t$   $\alpha_t$   $T_p(S)$   $\tilde{\alpha}_t$   $\exp_p^{-1}(p) = 0$   $\exp_p^{-1}(q) = \tilde{q}$   $\exp_p \circ \tilde{\alpha}_t = \alpha_t$

(c) (b)  $t \in [0, 1]$   $\epsilon > 0$   $t(\epsilon)$   $\alpha_{t(\epsilon)}$   $\tilde{\alpha}_{t(\epsilon)}$   $\tilde{\alpha}_{t(\epsilon)}$   $B$   
 $\epsilon$

$$l(\gamma_0) + l(\alpha_{t(\epsilon)}) \geq \frac{2\pi}{\sqrt{K_0}} - 2\epsilon.$$

(d) (c)  $\epsilon_n \rightarrow 0$   $\{t(\epsilon_n)\}$   $\alpha_{t_0}$   $t_0 \in [0, 1]$

$$l(\gamma_0) + l(\alpha_{t_0}) \geq \frac{2\pi}{\sqrt{K_0}}.$$

: Klingenberg  $2\pi/\sqrt{K_0}$   $\exp_p$   
 : (a) 5-5 3  $K \leq K_0$   $T_p(S)$   $\pi/\sqrt{K_0}$   $\exp_p$  — Jacobi

(b)  $\exp_p : B \rightarrow S$   $\exp_p|_B : B \rightarrow S$   $t$   $\alpha_t$   $\exp_p$   
 (c)  $t$   $0$   $\tilde{q}$   $T_p(S)$   $\tilde{\alpha}$   $\exp_p$   $\alpha$   $\tilde{\alpha}$   $l(\tilde{\alpha}) \geq |\tilde{q}|$  Gauss

$l(\alpha) \geq l(\gamma_0)$   $t \rightarrow 1$   $\tilde{q}$   $\tilde{q}$   $B$   $|\tilde{q}| = \pi/\sqrt{K_0}$   $l(\gamma_0) < \pi/\sqrt{K_0}$   
 $t < 1$   $t(\epsilon)$   $|\tilde{\alpha}_{t(\epsilon)}| > \pi/\sqrt{K_0} - \epsilon$   $l(\alpha_{t(\epsilon)}) > \pi/\sqrt{K_0} - \epsilon$

$$l(\gamma_0) + l(\alpha_{t(\epsilon)}) > \frac{2\pi}{\sqrt{K_0}} - \epsilon - 2\epsilon$$

(d)  $\epsilon_n \rightarrow 0$   $t_n = t(\epsilon_n)$   $t_{n_k} \rightarrow t_0$   $l(\alpha_{t_0}) \geq \limsup l(\alpha_{t_{n_k}})$   $l(\gamma_0) + l(\alpha_{t_0}) \geq 2\pi/\sqrt{K_0}$

5-6, 9a-b — do Carmo, Exercise 5-6, 9 a Klingenberg  $S$   $K \geq 0$   
 $\exp_p : T_p(S) \rightarrow S$  b (a) Hadamard : Klingenberg  
 $\pi/\sqrt{K_0}$

: (a)  $\exp_p(u_1) = \exp_p(u_2) = q$   $u_1 \neq u_2$   $\gamma_1, \gamma_2$   $l(\gamma_1) \leq l(\gamma_2)$   $\gamma_1$   
 $\gamma_2$   $l(\gamma_1) < \pi/\sqrt{K_0}$  Klingenberg  $l(\gamma_1) + l(\alpha_{t_0}) \geq 2\pi/\sqrt{K_0}$   $l(\gamma_1) <$   
 $\pi/\sqrt{K_0}$   $l(\alpha_{t_0}) \leq l(\gamma_1)$   $l(\gamma_1) \geq \pi/\sqrt{K_0}$   $l(\gamma_2) \geq \pi/\sqrt{K_0}$  Hopf-  
 Rinow  $d(p, q)$   $d(p, q) < \pi/\sqrt{K_0}$   $u_1 \neq u_2$   
 (b) (a)  $K \leq 0$   $K_0 = 0$   $\pi/\sqrt{K_0} \rightarrow \infty$   $\exp_p$   $\exp_p$   $K \leq 0$   
 $\exp_p$   
 5-6, 10\* Synge — do Carmo, Exercise 5-6, 10  $S$   $\alpha : [0, l] \rightarrow S$   
 $\alpha$   $0$   $l$   $\alpha_0, \alpha_1 : [0, l] \rightarrow S$   $H : [0, l] \times [0, 1] \rightarrow S$   
 $H(s, 0) = \alpha_0(s)$   $H(s, 1) = \alpha_1(s)$   $s \in [0, l]$   $H$   $\alpha_0$   $\alpha_1$   
 $S$   $K > 0$   $S$  : Synge  $K > 0$  " "  
 :  $\gamma$   $l$   $\gamma$   $n(s)$   $h(s, t) = \exp_{\gamma(s)}(tn(s))$   $t \in (-\epsilon, \epsilon)$   
 $L(t)$   $L'(0) = 0$   $\gamma$

$$L''(0) = \int_0^l (|Dn/ds|^2 - K) ds.$$

$n$   $|Dn/ds| = 0$   $K > 0$   $L''(0) < 0$   $t \neq 0$   $L(t) < L(0)$   $h(\cdot, t)$   
 $h(0, t) = h(l, t)$   
 5-6, 11 — do Carmo, Exercise 5-6, 11  $S$   $p$   $S$   $p$   
 Klingenberg  $\exp_p : T_p(S) \rightarrow S$  :  $\exp_p$  Jacobi  $\exp_p$   
 1  
 :  $\exp_p$   $T_p(S)$   $u_1 \neq u_2$   $\exp_p(u_1) = \exp_p(u_2) = q$   $\gamma_1, \gamma_2$   
 Klingenberg  $K_0$   $K$   $t_0$   $l(\gamma_1) + l(\alpha_{t_0}) \geq 2\pi/\sqrt{K_0}$   
 $\pi/\sqrt{K_0} \rightarrow \infty$   $K_0$   $l(\gamma_1)$   $\exp_p$   $+$   $=$   
 5-9, 1 — Bonnet  $S$   $K \geq \delta > 0$  5-5 5  $\gamma : [0, \infty) \rightarrow$   
 $S$   $(0, \pi/\sqrt{\delta})$   $\gamma(0)$  Jacobi  $S$   $diam(S) \leq \pi/\sqrt{\delta}$  5-4  
 Bonnet : Jacobi  $K \geq \delta$   $s = \pi/\sqrt{\delta}$  —  
 $\pi/\sqrt{\delta}$   
 :  $p, q \in S$   $\gamma$  Jacobi  $\gamma$  5-5 5  $(0, \pi/\sqrt{\delta}]$   $\gamma$   
 $l(\gamma) \leq \pi/\sqrt{\delta}$   $d(p, q) \leq \pi/\sqrt{\delta}$   $diam(S) \leq \pi/\sqrt{\delta}$   $\Rightarrow S$   
 5-9, 2 —  $\gamma : (-\infty, \infty) \rightarrow S$  line

1.  $x^2 + y^2 = 1$

2.  $S$   $K > 0$   $\gamma : (-\infty, \infty) \rightarrow S$   $S$   $J(s)$   $\gamma$  Jacobi  
 $\langle J(0), \gamma'(0) \rangle = 0$   $|J(0)| = 1$   $J'(0) = 0$   $J(s) = u(s)e_2(s)$   $u$   $u'' + Ku =$   
 $0$   $u(0) = 1$   $u'(0) = 0$

3.  $K > 0$   $\epsilon > 0$   $u(\epsilon) > 0$   $u(-\epsilon) > 0$   $u'(\epsilon) < 0$   $u'(-\epsilon) > 0$   $s_0$   
 $J(s)$   $(-s_0, s_0)$

4.

: Jacobi —  $u(s) = \cos(s)$  Jacobi " "  
 : (1)  $x^2 + y^2 = 1$   
 (2) 5-5 1  $J$   $\gamma'$   $J(s) = u(s)e_2(s)$  Jacobi  $u'' + Ku = 0$   $J(0) =$   
 $e_2(0)$   $J'(0) = 0$   $u(0) = 1$   $u'(0) = 0$



(3)  $K > 0$   $u$   $v'' = 0$   $K = 0$   $K > 0$   $u$   $\epsilon$   $v$   $u'(\epsilon) < v'(\epsilon) <$   
 $0$   $u'(-\epsilon) > w'(-\epsilon) > 0$   $u$   $J$   $(-s_0, s_0)$   
 (4)  $\gamma$   $S$   $s$   $d(\gamma(s), p) = |s|$  (3)  $J$   $(-\infty, \infty)$   $\gamma$   
 $3-3$   $K = 0$   $K > 0$  /  $K \equiv 0$  /  $K < 0$  /  $K = 0$   
 ??  $K = 0$   
 2016 A. V. Pogorelov (1956) Pogorelov  
 Mainardi-Codazzi  $K = 0$  4-3  
 $K \equiv 0$  " "  $R^3$   $R^3$   
 $n$   
 Mainardi-Codazzi  
 5-3 5-4 5-5 5-6 A 5-7 5-9 5-11  
 5-8 5-11 Hilbert  $R^3$  Hilbert  $K < 0$   $K \equiv 0$   
 $S$   $K = 0$   $H = 0$   $S$   
 " "  $R^3$  5-10  
 $K = 0$   $R^3$   
 translation surfaces  
 Mainardi-Codazzi  $K = 0$   $(e_v = \frac{E_v}{2}(k_1 + k_2))$   
 $K = 0$   
 $K = 0$   $R^3$  5-2  
 $K = 0$   $R^3$  Gauss  
 5-2 5-6 — Hopf-Rinow Bonnet Jacobi Hadamard —  
 5-8  
 Jacobi  
 5-10  $R^3$   
 5-11 Hilbert " "  $R^3$   
 5

## 1.9 Jacobi's Theorems

Jacobi  
 [Jacobi]  $\gamma : [0, l] \rightarrow S$   $S$   $\gamma$   
 $\cdot$   $\gamma$   $V(s)$   $V(0) = V(l) = 0$

$$L''(0) = \int_0^l (|DV/ds|^2 - K|V|^2) ds.$$

Sturm-Liouville  $V \equiv 0$  21  
 $\cdot$   $\gamma$   $q = \gamma(s_0)$  Jacobi  $J(s)$   $J(0) = J(s_0) = 0$  Jacobi  
 $L''(0) < 0$   $\gamma$

---

<sup>21</sup> sec:derivation-jacobi-minimality

Jacobi " " focal caustics

Jacobi

$$K \leq 0 \quad ?? \quad K \leq 0$$

Jacobi 5-5 Jacobi

Morse Morse

Jacobi 5-4 5-5 Jacobi

## 1.10 Abstract Surfaces; Further Generalizations

$R^3$   
[ ] abstract surface  $M$   $g$   $(M, g)$  Riemannian  
surface

$$R^3 \quad R^3 \quad R^3 \quad R^3$$

. 4 4-2 4-7

• 4-2

• Gauss-Bonnet 4-5 — -

• Hopf-Rinow  $\Rightarrow$

• Hadamard  $+ K \leq 0 + \Rightarrow R^2$

• Bonnet  $+ K \geq \delta > 0 \Rightarrow$

.  $n$   $(M, g)$

•  $K$   $(0, 4)$  Ricci sectional

• Jacobi  $n$

• Hopf-Rinow Bonnet Hadamard

• Ricci  $\Rightarrow$  Bonnet

.  $R^3$  Nash Nash  $R^3$   
Hilbert  $K < 0$

" "

$R^3$   
Hadamard Bonnet



isometry. What is the (Gaussian) curvature of such a metric? :  $P^2$   
 $S^2$

$$\begin{array}{ccccccc} : & \pi : S^2 \rightarrow P^2 & S^2 & g_0 & K = 1 & P^2 & g \\ g|_U = (\pi|_V)^{-1*} g_0|_V & V = \pi^{-1}(U) & \pi & & \pi & P^2 & S^2 \\ K = 1 & & & & & & \end{array}$$

5-10, 2 — do Carmo, Exercise 5-10, 2 (The Infinite Möbius Strip.) Let

$$C = \{(x, y, z) \in R^3; x^2 + y^2 = 1\}$$

be a cylinder and  $A: C \rightarrow C$  be the map (the antipodal map)  $A(x, y, z) = (-x, -y, -z)$ . Let  $M$  be the quotient of  $C$  by the equivalence relation  $p \sim A(p)$ , and let  $\pi: C \rightarrow M$  be the map  $\pi(p) = \{p, A(p)\}$ ,  $p \in C$ .

a. Show that  $M$  can be given a differentiable structure so that  $\pi$  is a local diffeomorphism ( $M$  is then called the infinite Möbius strip).

b. Prove that  $M$  is nonorientable.

c. Introduce on  $M$  a Riemannian metric so that  $\pi$  is a local isometry. What is the curvature of such a metric? :  $C$   $A$

$$\begin{array}{ccccccc} : & (a) & C & A & M = C/\sim & A^2 = \text{id} & \pi : C \rightarrow M \\ & & \pi & & & & M \end{array}$$

$$\begin{array}{ccccccc} (b) & M & C & (u, v) & u & v & A \\ & & & & & & (u, v) \mapsto (u + \pi, -v) \\ d(u + \pi) \wedge d(-v) = -du \wedge dv & & & & & & du \wedge dv \\ & & & & & & M \end{array}$$

$$\begin{array}{ccccccc} (c) & M & g & C & A & A & g \\ & & & & & & M \\ & & & & & & \pi \\ & & & & & & C \\ & & & & & & 0 \\ & & & & & & M \\ & & & & & & 0 \end{array}$$

5-10, 3 — do Carmo, Exercise 5-10, 3 a. Show that the projection  $\pi: S^2 \rightarrow P^2$  from the sphere onto the projective plane has the following properties: (1)  $\pi$  is continuous and  $\pi(S^2) = P^2$ ; (2) each point  $p \in P^2$  has a neighborhood  $U$  such that  $\pi^{-1}(U) = V_1 \cup V_2$ , where  $V_1$  and  $V_2$  are disjoint open subsets of  $S^2$ , and the restriction of  $\pi$  to each  $V_i$ ,  $i = 1, 2$ , is a homeomorphism onto  $U$ . Thus,  $\pi$  satisfies formally the conditions for a covering map (see Sec. 5-6, Def. 1) with two sheets. Because of this, we say that  $S^2$  is an orientable double covering of  $P^2$ .

b. Show that, in this sense, the torus  $T$  is an orientable double covering of the Klein bottle  $K$  (cf. Example 2) and that the cylinder is an orientable double covering of the infinite Möbius strip (cf. Exercise 2).

torus Klein

$$\begin{array}{ccccccc} : & (a) & \pi : S^2 \rightarrow P^2 & (p, -p) & U \subset P^2 & \pi^{-1}(U) & V_1, V_2 \\ U & & & & & & \end{array}$$

$$\begin{array}{ccccccc} (b) & \text{torus } T & \text{Klein } K & 2 & T = R^2/Z^2 & K = R^2/(x, y) \sim (x + 1, -y) \\ (x, y) \mapsto (x, y) & & & & C = R^2/(x, y) \sim (x + 1, y) & M = R^2/(x, y) \sim \\ (x + 1, -y) & & & & & & \end{array}$$

5-10, 4 — do Carmo, Exercise 5-10, 4 (The Orientable Double Covering.)

This exercise gives a general construction for the orientable double covering of a nonorientable surface. Let  $S$  be an abstract, connected, nonorientable surface. For each  $p \in S$ , consider the set  $B$  of all bases of  $T_p(S)$  and call two bases equivalent if they are related by a matrix with positive determinant. This is clearly an equivalence relation and divides  $B$  into two disjoint sets (cf. Sec.

1-4). Let  $D_p$  be the quotient space of  $B$  by this equivalence relation.  $D_p$  has two elements, and each element  $O_p \in D_p$  is an orientation of  $T_p(S)$  (cf. Sec. 1-4). Let  $\tilde{S}$  be the set

$$\tilde{S} = \{(p, O_p); p \in S; O_p \in D_p\}.$$

To give  $\tilde{S}$  a differentiable structure, let  $\{U_\alpha, \mathbf{x}_\alpha\}$  be the maximal differentiable structure of  $S$  and define  $\tilde{\mathbf{x}}_\alpha: U_\alpha \rightarrow \tilde{S}$  by

$$\tilde{\mathbf{x}}_\alpha(u_\alpha, v_\alpha) = \left( \mathbf{x}_\alpha(u_\alpha, v_\alpha), \left[ \frac{\partial}{\partial u_\alpha}, \frac{\partial}{\partial v_\alpha} \right] \right),$$

where  $(u_\alpha, v_\alpha) \in U_\alpha$  and  $[\partial/\partial u_\alpha, \partial/\partial v_\alpha]$  denotes the element of  $D_p$  determined by the basis  $\{\partial/\partial u_\alpha, \partial/\partial v_\alpha\}$ . Show that

a.  $\{U_\alpha, \tilde{\mathbf{x}}_\alpha\}$  is a differentiable structure on  $\tilde{S}$  and that  $\tilde{S}$  with such a differentiable structure is an orientable surface.

b. The map  $\pi: \tilde{S} \rightarrow S$  given by  $\pi(p, O_p) = p$  is a differentiable surjective map. Furthermore, each point  $p \in S$  has a neighborhood  $U$  such that  $\pi^{-1}(U) = V_1 \cup V_2$ , where  $V_1$  and  $V_2$  are disjoint open subsets of  $\tilde{S}$  and  $\pi$  restricted to each  $V_i$ ,  $i = 1, 2$ , is a diffeomorphism onto  $U$ . Because of this,  $\tilde{S}$  is called an orientable double covering of  $S$ .

$$\begin{aligned} & \begin{matrix} V_1, V_2 \\ \{ \tilde{\mathbf{x}}_\alpha \} \end{matrix} : \tilde{S} \quad \begin{matrix} \tilde{\mathbf{x}}_\alpha(u, v) = \tilde{\mathbf{x}}_\beta(w, z) \\ \{ \tilde{\mathbf{x}}_\alpha^{-1} \circ \tilde{\mathbf{x}}_\beta \} \end{matrix} \quad \begin{matrix} \mathbf{x}_\alpha(u, v) = \mathbf{x}_\beta(w, z) \\ \{ \mathbf{x}_\beta^{-1} \circ \mathbf{x}_\alpha \} \end{matrix} \\ & \tilde{p} = (p, O_p) \in \tilde{S} \quad \begin{matrix} O_p \\ T_p(S) \end{matrix} \quad \begin{matrix} U \ni p \\ U \subset S \end{matrix} \quad \begin{matrix} \mathbf{x} : U \rightarrow R^2 \\ V = \tilde{\mathbf{x}}(U) \subset \tilde{S} \end{matrix} \end{aligned}$$

$$\tilde{\mathbf{x}}(u, v) = (p, O_p), \quad p = \mathbf{x}(u, v),$$

$$\begin{aligned} & \begin{matrix} O_p \\ \pi \end{matrix} : \begin{matrix} \{ \partial/\partial u, \partial/\partial v \} \\ p \in S \end{matrix} \quad \begin{matrix} U \\ U \subset S \end{matrix} \quad \begin{matrix} V \\ \exp_p \end{matrix} : \begin{matrix} \tilde{\mathbf{x}} \\ B_\delta \subset T_p(S) \end{matrix} \quad \begin{matrix} \tilde{S} \\ U \end{matrix} \quad \begin{matrix} D_q \\ q \in U \end{matrix} \end{aligned}$$

$$V_1 = \{(q, O_q) : q \in U, O_q T_q(S)\}, \quad V_2 = \{(q, O_q) : q \in U, O_q\}.$$

$$V_1, V_2 \subset \tilde{S} \quad \pi^{-1}(U) = V_1 \cup V_2 \quad \pi|_{V_i} : V_i \rightarrow U \quad q \mapsto (q, O_q)$$

5-10, 5 — do Carmo, Exercise 5-10, 5 Extend the Gauss-Bonnet theorem (see Sec. 4-5) to orientable geometric surfaces and apply it to prove the following facts:

a. There is no Riemannian metric on an abstract surface  $T$  diffeomorphic to a torus such that its curvature is positive (or negative) at all points of  $T$ .

b. Let  $T$  and  $S^2$  be abstract surfaces diffeomorphic to the torus and the sphere, respectively, and let  $\varphi: T \rightarrow S^2$  be a differentiable map. Then  $\varphi$  has at least one critical point, i.e., a point  $p \in T$  such that  $\det(d\varphi_p) = 0$ . : Gauss-Bonnet

$$\begin{aligned} & \chi = 0 \quad K > 0 \quad K < 0 \quad \int K dA \quad K < 0 \\ & : (a) \quad T \quad \text{Gauss-Bonnet} \quad \int_T K dA = 2\pi\chi(T) = 0 \quad K > 0 \quad \int K dA > 0 \\ & 0 \quad K < 0 \quad \int K dA < 0 \end{aligned}$$

(b)  $\varphi : T \rightarrow S^2$   $p \in S^2$   $\det(d\varphi_p) \neq 0$   $p \in \varphi(T)$   $D \subset \partial D$   $S^2$   
 $\varphi|_D : D \rightarrow S^2$  — Sard  $\varphi$

5-10, 6 — do Carmo, Exercise 5-10, 6 Consider the upper half-plane  $R_+^2$  (cf. Example 3) with the metric

$$E(x, y) = 1, \quad F(x, y) = 0, \quad G(x, y) = \frac{1}{y}, \quad (x, y) \in R_+^2.$$

Show that the lengths of vectors become arbitrarily large as we approach the boundary of  $R_+^2$  and yet the length of the vertical segment

$$x = 0, \quad 0 < \epsilon \leq y \leq 1,$$

approaches 2 as  $\epsilon \rightarrow 0$ . Conclude that such a metric is not complete.

$$y \rightarrow 0^+ \quad G = 1/y \rightarrow \infty$$

$$: p = (x_0, y_0) \in R_+^2 \quad v = (v_1, v_2)$$

$$|\mathbf{v}| = \sqrt{Ev_1^2 + 2Fv_1v_2 + Gv_2^2} = \sqrt{v_1^2 + v_2^2/y} \geq \frac{|v_2|}{\sqrt{y}}.$$

$$y \rightarrow 0^+ \quad v_2 \neq 0 \quad |\mathbf{v}| \rightarrow \infty \quad x = 0 \quad y \in [\epsilon, 1]$$

$$L = \int_{\epsilon}^1 \sqrt{G} dy = \int_{\epsilon}^1 \frac{1}{\sqrt{y}} dy = 2(1 - \sqrt{\epsilon}) \rightarrow 2 \quad (\epsilon \rightarrow 0).$$

$\alpha_n(t) = (0, 1/n)$   $n \rightarrow \infty$   $\alpha_n$   $n, m$   $d(\alpha_n, \alpha_m) \leq d(\alpha_n, \alpha_m)_{\text{euclidean}} \rightarrow 0$   
 $1/n \rightarrow 0$   $\{\alpha_n\}$  Cauchy  $\alpha_n$   $y = 0$   $S$   $S$  Cauchy —

5-10, 7 — do Carmo, Exercise 5-10, 7 Prove that the Poincaré half-plane (cf. Example 3) is a complete geometric surface. Conclude that the hyperbolic plane is complete.

$$y \rightarrow 0 \quad : \quad |\mathbf{v}| = \sqrt{Ev_1^2 + 2Fv_1v_2 + Gv_2^2} = \sqrt{v_1^2 + v_2^2/y} \quad y \rightarrow 0^+ \quad v_2 \neq 0$$

$$0 \quad |\mathbf{v}| \rightarrow \infty \quad x = 0 \quad y \in [\epsilon, 1]$$

$$L = \int_{\epsilon}^1 \sqrt{G} dy = \int_{\epsilon}^1 \frac{1}{\sqrt{y}} dy = 2(1 - \sqrt{\epsilon}) \rightarrow 2 \quad (\epsilon \rightarrow 0).$$

$\alpha_n(t) = (0, 1/n)$   $n \rightarrow \infty$   $y = 0$  Cauchy  $\alpha_n$   $S$   
 $S$

5-10, 8 — do Carmo, Exercise 5-10, 8 Another way of finding the geodesics of the Poincaré half-plane (cf. Example 3) is to use the Euler-Lagrange equation for the corresponding variational problem (cf. Exercise 4, Sec. 5-4). Since we know that the vertical lines are geodesics, we can restrict ourselves to geodesics of the form  $y = y(x)$ . Thus, we must look for the critical points of the integral  $F = 0$

$$\int \sqrt{E + G(y')^2} dx = \int \sqrt{\frac{1 + (y')^2}{y}} dx,$$

since  $E = G = 1/y^2$ . Use Exercise 4, Sec. 5-4, to show that the solution to this variational problem is a family of circles of the form

$$(x + k_1)^2 + y^2 = k_2^2, \quad k_1, k_2 = \text{const.}$$

$$\begin{array}{llll} \text{Poincaré} & ds^2 = (dx^2 + dy^2)/y^2 & y \rightarrow 0^+ & \\ \text{Poincaré} & ds^2 = (dx^2 + dy^2)/y^2 & 6 \text{ Cauchy} & y \rightarrow 0 \\ y \geq \epsilon > 0 & \text{Hopf-Rinow} & \text{Poincaré} & \end{array}$$

5-10, 9 — do Carmo, Exercise 5-10, 9 Let  $\tilde{S}$  and  $S$  be connected geometric surfaces and let  $\pi: \tilde{S} \rightarrow S$  be a surjective differentiable map with the following property: For each  $p \in S$ , there exists a neighborhood  $U$  of  $p$  such that  $\pi^{-1}(U) = \bigcup_{\alpha} V_{\alpha}$ , where the  $V_{\alpha}$ 's are open disjoint subsets of  $\tilde{S}$  and  $\pi$  restricted to each  $V_{\alpha}$  is an isometry onto  $U$  (thus,  $\pi$  is essentially a covering map and a local isometry).

a. Prove that  $S$  is complete if and only if  $\tilde{S}$  is complete.  
 b. Is the metric on the infinite Möbius strip, introduced in Exercise 2, part c, a complete metric? : ——— Cauchy  
 : (a)  $S \quad \tilde{p}_n \quad \tilde{S} \quad \text{Cauchy} \quad \pi(\tilde{p}_n) \quad S \quad \text{Cauchy} \quad S \quad q \in S \quad q$   
 $U \quad \pi^{-1}(U) = \bigcup V_{\alpha} \quad n \quad \pi(\tilde{p}_n) \in U \quad \tilde{p}_n \quad V_{\alpha} \quad \pi|_{V_{\alpha}} \quad \tilde{p}_n \quad V_{\alpha}$

(b)  
 5-10, 10 — do Carmo, Exercise 5-10, 10 (Kazdan-Warner's Results.)  
 a. Let a metric on  $R^2$  be given by

$$E(x, y) = 1, \quad F(x, y) = 0, \quad G(x, y) > 0, \quad (x, y) \in R^2.$$

Show that the curvature of this metric is given by

$$\frac{\partial^2(\sqrt{G})}{\partial x^2} + K(x, y)\sqrt{G} = 0.*$$

b. Conversely, given a function  $K(x, y)$  on  $R^2$ , regard  $y$  as a parameter and let  $\sqrt{G}$  be the solution of (\*) with the initial conditions

$$\sqrt{G}(x_0, y) = 1, \quad \frac{\partial \sqrt{G}}{\partial x}(x_0, y) = 0.$$

Prove that  $G$  is positive in a neighborhood of  $(x_0, y)$  and thus defines a metric in this neighborhood. This shows that every differentiable function is locally the curvature of some (abstract) metric.

\*c. Assume that  $K(x, y) \leq 0$  for all  $(x, y) \in R^2$ . Show that the solution of part b satisfies

$$\sqrt{G(x, y)} \geq \sqrt{G(x_0, y)} = 1 \quad \text{for all } x.$$

Thus,  $G(x, y)$  defines a metric on all of  $R^2$ . Prove also that this metric is complete. This shows that any nonpositive differentiable function on  $\mathbf{R}^2$  is the curvature of some complete metric on  $\mathbf{R}^2$ . If we do not insist on the metric being complete, the result is true for any differentiable function  $K$  on  $R^2$ . Compare J. Kazdan and F. Warner, "Curvature Functions for Open 2-Manifolds," Ann. of

Math. 99 (1974), 203-219, where it is also proved that the condition on  $K$  given in Exercise 2 of Sec. 5-4 is necessary and sufficient for the metric to be complete.

$$\begin{aligned}
 & : \text{Kazdan-Warner} \\
 & : (a) \quad (x, y) \quad K = -\frac{1}{\sqrt{EG}} \left( \frac{\partial}{\partial x} \left( \frac{G_x}{\sqrt{EG}} \right) + \frac{\partial}{\partial y} \left( \frac{E_y}{\sqrt{EG}} \right) \right) \quad E = 1 \quad F = 0 \\
 & K = -\frac{1}{\sqrt{G}} \frac{\partial^2 \sqrt{G}}{\partial x^2} \quad (*) \\
 & (b) \quad K \quad (*) \quad \sqrt{G} \quad \text{ODE} \quad \sqrt{G}(x_0, y) = 1 \quad (\sqrt{G})_x(x_0, y) = 0 \\
 & \sqrt{G} \quad x_0 \quad \sqrt{G} > 0 \quad G > 0 \\
 & (c) \quad K \leq 0 \quad (\sqrt{G})_{xx} = -K\sqrt{G} \geq 0 \quad \sqrt{G} \quad x \quad \sqrt{G}(x_0) = 1 \quad (\sqrt{G})_x(x_0) = 0 \\
 & 0 \quad \sqrt{G}(x) \geq 1 \quad x \quad G > 0 \quad R^2 \quad G \geq 1 \quad \text{Hopf-Rinow}
 \end{aligned}$$

## 5-11

$$\begin{aligned}
 & 5-11, 1 \text{ — Stoke Remark Efimov} \quad S \quad K \leq \delta < 0 \quad \varphi: S \rightarrow R^3 \\
 & H \quad \text{Remark 2} \quad \text{Efimov} \\
 & a. \quad \varphi \quad \text{Gauss} \quad N: \varphi(S) \subset R^3 \rightarrow S^2 \quad S^2 \quad K \neq 0 \quad N \quad S \\
 & (,) \quad N \circ \varphi: S \rightarrow S^2 \quad S \quad \varphi \quad \varphi(S)
 \end{aligned}$$

$$g_{11} = (k_1)^2 E, \quad g_{12} = 0, \quad g_{22} = (k_2)^2 G,$$

$$E, F(=0), G$$

$$b. \quad M > 0 \quad k_1^2 < M \quad k_2^2 < M$$

$$c. \quad b \quad S \quad : \text{Efimov} \quad \text{Hilbert}$$

Gauss

$$\begin{aligned}
 & : (a) \quad \text{Gauss} \quad N: S \rightarrow S^2 \quad N \circ \varphi \quad g_{11} = (k_1)^2 E \quad g_{22} = \\
 & (k_2)^2 G \quad g_{12} = 0
 \end{aligned}$$

$$\begin{aligned}
 & (b) \quad K = k_1 k_2 \neq 0 \quad |k_i| \leq M \quad H \quad |H| \leq C \quad |K| \geq k_0^2 > 0 \\
 & |k_i| \leq C/k_0
 \end{aligned}$$

$$\begin{aligned}
 & (c) \quad (b) \quad \tilde{S} \quad |K| \geq k_0^2 > 0 \quad K' = 1 \quad \text{Bonnet} \quad \tilde{S} \quad \tilde{S} \quad S \quad N \circ \varphi \quad S \\
 & \tilde{S} \quad S \quad S \quad ?? \quad > 0 \quad K < 0
 \end{aligned}$$

$$\begin{aligned}
 & 5-11, 2 \text{ — Efimov} \quad K \leq \delta < 0 \quad S \subset R^3 \quad \text{Efimov} \\
 & S \subset R^3
 \end{aligned}$$

$$\begin{aligned}
 & a. \quad S \quad 5-58(a) \quad (b) \quad 5-58(b) \quad z \\
 & b. \quad (\varphi(s), \psi(s)) \quad s \in R \quad \psi(0) = 0 \quad \varphi'' + K\varphi = 0 \quad 3-3 \quad 4 \quad (9)
 \end{aligned}$$

$$K \leq \delta < 0 \quad s_0 \in [0, +\infty) \quad (\varphi'(s_0))^2 = 1$$

$$c. \quad p_0 = (\varphi(s_0), \psi(s_0)) \quad (\varphi(s), \psi(s)) \quad 5-58(c) \quad \text{I II III}$$

S

[Figure 5-58(a)][width=0.45]../../PDFs/differential-geometry/transcript/Do

Carmo - Differential Geometry of Curves and Surfaces/hybridocr/images/98f40d772f751b0834697b4

58(b)][width=0.45]../../PDFs/differential-geometry/transcript/DoCarmo - Differential

[Figure 5-58(c)][width=0.6]../../PDFs/differential-geometry/transcript/Do

Carmo - Differential Geometry of Curves and Surfaces/hybridocr/images/6e3e016f2d3b4b0520d5161b

: Efimov

$$\varphi'' + K\varphi = 0 \quad K < 0 \quad \varphi$$

$$: (a) \quad 3-3 \quad K = -\varphi''/\varphi \quad K < 0 \quad \varphi'' > 0 \quad \varphi \quad \psi(0) =$$

$$\psi(l) = 0 \quad \varphi > 0 \quad \text{fig:fig5-58a} \quad (b)$$



- (b)  $s \in [0, +\infty)$   $K \leq \delta < 0$   $\varphi'' + \delta\varphi \leq 0$   $\varphi'(0) > 0$   $\varphi' \quad s_0$   
 $\varphi'(s_0)^2 = 1$   $\varphi \quad \varphi \rightarrow \infty$   $K$
- (c)  $p_0 \quad \varphi(s)$  I II III I  $\varphi$   $S$  II III  $\varphi$   $K$   
 $K \leq \delta < 0$   $S$
- 5-11, 3\* — T.K. Milnor Hilbert  $S$   $g_1$   $K \equiv -1$   $\varphi: S \rightarrow$   
 $R^3$
- a. Gauss  $N: \varphi(S) \subset R^3 \rightarrow S^2$   $g_2$   $S$   $N \circ \varphi: S \rightarrow S^2$   $S$   
 $\varphi \quad \varphi(S)$   $g_1$

$$du^2 + 2 \cos \theta \, dudv + dv^2,$$

$g_2$

$$du^2 - 2 \cos \theta \, dudv + dv^2.$$

- b.  $g_3 = \frac{1}{2}(g_1 + g_2)$   $S$   $g_1$   $3g_3 \geq g_1$   $g_3$   
c.  $g_3$   $R^2$   $\psi: S \rightarrow R^2$   $\psi$   $S$   $R^2$
- d. c  $S$  Hilbert : Milnor Gauss  $g_3 =$   
 $(g_1 + g_2)/2$   $K = 0$  Hilbert  
: (a)  $(u, v)$   $g_1 = du^2 + 2 \cos \theta \, dudv + dv^2$   $g_2 = du^2 - 2 \cos \theta \, dudv + dv^2$   
 $\theta$
- (b)  $g_3 = (g_1 + g_2)/2 = du^2 + dv^2$   $K(g_3) = 0$   $3g_3 = g_1 + 2g_3 \geq g_1$
- $g_1$   $g_3$   
(c)  $K = 0$   $R^2$  (b)  $\psi: S \rightarrow R^2$   $g_1$   $(1, \pm 1)$   $R^2$   $\pm 1$
- (d)  $\psi$   $S$   $g_1$   $R^2$  Hilbert " " —  $S$



# Appendix A

## Euclidean

5

### A.1

$$\begin{array}{l} A \subset \mathbb{R}^n \quad \text{connected} \\ A \subset \mathbb{R}^n \quad \text{compact} \end{array} \quad \Leftrightarrow \quad \text{Heine-Borel}$$

$$\begin{array}{l} \mathbb{R}^3 \\ A \subset \mathbb{R}^n \quad p \in \mathbb{R}^n \quad A \quad \text{boundary point} \quad p \quad A \quad \mathbb{R}^n - A \quad A \\ \text{Bd } A \\ S \subset \mathbb{R}^3 \quad \mathbb{R}^3 \quad \mathbb{R}^3 \end{array}$$

### A.2

$$\begin{array}{l} B \subset \mathbb{R}^n \quad \text{arcwise connected} \quad p, q \in B \quad \alpha : [0, 1] \rightarrow B \quad \alpha(0) = \\ p \quad \alpha(1) = q \end{array}$$

$$\begin{array}{l} \mathbb{R}^n \\ \mathbb{R}^3 \end{array}$$

### A.3

$$\begin{array}{l} \alpha_0, \alpha_1 : [0, l] \rightarrow B \quad \text{homotopic} \quad H : [0, l] \times [0, 1] \rightarrow B \\ H(s, 0) = \alpha_0(s) \quad H(s, 1) = \alpha_1(s) \quad H(0, t) \quad H(l, t) \quad t \\ B \quad \text{simply connected} \\ S^2 \\ \mathbb{R}^2 \end{array}$$

A.4 Cauchy

$A \subset \mathbb{R}^n$     $\{p_k\}$  **Cauchy**    $\epsilon > 0$     $N$     $k, l \geq N$     $|p_k - p_l| < \epsilon$   
Cauchy  
 $\mathbb{R}^n$    Cauchy

A.5

5

- $S \Rightarrow$
- 5-3
- Hadamard 5-6
- Cauchy Hopf-Rinow

Munkres Topology Milnor Topology from  
the Differential Viewpoint